

*Departure From the Photon Baseline:
A Universal Ruler for Mass, Gravity, and Time*

The Three-Axis Framework and the Nine-State Classification of the 27 Geometric States

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Paper 2 of the Series

Abstract

This paper establishes the geometric foundation of the framework introduced in this series. A configuration in the framework is a structure of three triangles meeting at a central junction. Each triangle holds three integer values — one at the central junction where the three triangles meet, and two at its outer corners. Across the three triangles, a configuration occupies nine positions in total, with content always balanced as three X values, three Y values, and three O values. The framework's central physical commitment is that the photon configuration propagates at the speed of light because it carries zero tilt; everything that carries mass is a configuration whose triangles have been twisted into a self-contained resolved structure through axis-tilt operations, and the resulting tilt cost is what slows the configuration below c . Three rotational axes with three discrete values per axis produce exactly twenty-seven configurations, partitioned into nine categories by central junction content signature and designated States 1 through 9. Two of these categories are self-contained configurations capable of stable, independent existence: State 9 (the photon) and State 1 (the electron-positron pair). The remaining seven categories are open configurations that achieve stability only by bonding into compositional structures. Bonding occurs when outer corners of one configuration touch the central junction of another, or when three matching outer corners from three different configurations converge at a single meeting point. Chains of bonding events progress through stages named by their geometry: a dual-axis-plus-one-surplus configuration (first link), a tri-axis-plus-one-surplus configuration (conventionally the meson), and a triple tri-axis configuration (resolution achieved, conventionally the baryon), with tri-axis-plus-two-surplus configurations (conventionally tetraquarks and heavier exotics) forming transiently when surplus exceeds what resolution can hold. Charge is reframed as the directional axial orientation a configuration takes when unwinding back toward the photon baseline. This framework replaces traditional force-carrier models by establishing that interaction dynamics are governed entirely by local metric screening and a coordinate destination-failure mechanism within the tri-axis coordinate manifold. One appendix follows the main text: Appendix B derives the 720° spin signature of mass configurations from the framework's geometric structure. The framework's critiques of the conventional interpretation of charge, the nuclear interior, the so-called neutrino, and collider spectra are developed in the companion critique paper, Paper 2.5.

1. Terms and Conventions

The following terms are defined once here and used throughout this paper and the series. Defining them up front lets the body refer to each mechanism by name rather than re-deriving it on every appearance.

- **Configuration** — a structure of three triangles meeting at a central junction, occupying nine positions (three central, six outer), with content always balanced as three X, three Y, and three O values.
- **X, Y, O** — the three discrete tilt values an axis can hold, called structural integers. They denote direction of tilt, not arithmetic quantity, and carry no notion of positive, negative, or charge. No value is privileged over another; X and Y are simply different directions, and O is the un-tilted anchor.
- a_k — the universal identifier, the structural punctuation mark that tells one discrete state, orientation, or configuration apart from the next. It is not a distance, a measurement, or an active process; it does not scan, monitor, or register. It is only the tag that marks each instance distinct on the ledger.
- **Identity scan** — the inner 360° sweep by which a configuration confirms what it is (photon baseline, resolved State 1, or open state).
- **Temporal scan** — the outer 360° sweep by which a configuration carrying tilt searches its local surroundings for resolution. The accumulation of these sweeps is what registers time; the photon, carrying no tilt, runs no temporal scan and registers no time.
- **Manifold scan** — the full 720° cycle of the identity scan plus the temporal scan, run by every configuration that carries mass.
- **Mass** — one phenomenon at three levels: the geometric departure of a configuration from the photon baseline (what it is), the accumulated tilt count in the nine-position ledger (how it is measured), and the activation of the temporal scan that slows the configuration below c (how it manifests).
- **A note on the names "proton" and "neutron":** this framework does not use them. They carry the assumption that a composite is a container of pre-packaged parts, which the framework rejects. The structures conventionally called the proton and neutron are named here by their geometry, and the geometric names are used throughout the series. The conventional name is given on first use only as a bridge for the reader.
- **Heavy X-dominant configuration** — a resolved three-node composite whose net tilt leans along the X-axis. ("Heavy" denotes a resolved-configuration-scale composite carrying substantial accumulated tilt; "dominant" denotes which axis its net tilt favors.) This is one of the two structures conventionally called the proton.
- **Heavy Y-dominant configuration** — the identical structure with its net tilt leaning along the Y-axis instead. X-dominant and Y-dominant are the *same composite* in opposite tilt directions, the way the electron and positron are one geometry spun opposite ways; neither direction is privileged, and there is no "positive" or "negative" involved. This is the opposite-tilt counterpart conventionally called the antiproton.
- **Heavy balanced configuration** — the distinct structure conventionally called the neutron. It is *not* a third tilt direction of the same composite; it is a different geometry entirely — three State 4 configurations merging at a single mutual center point — whose net tilt is globally balanced rather than axis-dominant.
- **States 1–9** — the nine categories of configuration by central-junction signature; State 9 is the photon, State 1 the electron-positron pair, States 2–8 the open states.

2. Purpose of This Paper

Paper 1 of this series surveyed twenty-seven published Lagrangian approaches to gravity and observed that every framework in the field adopts some structure as input and derives consequences from the adoption. That paper identified the framework introduced in this series as the twenty-seventh approach with a minimal adoption set: three rotational axes, three discrete values per axis, the photon baseline as the parent state at c , and one geometric principle of resolution.

This paper begins the framework's positive case. It establishes the geometric structure that organizes all subsequent papers in the series. The framework's primary classification organizes the twenty-seven possible configurations into nine categories designated States 1 through 9, each distinguished by the content at its central junction. Two categories are self-contained configurations stable on their own. The remaining seven are open configurations that achieve stability through bonding into compositional structures. Subsequent papers will use the geometric vocabulary established here to express the mechanics of excess-tilt reconciliation toward the photon state.

The presentation is structural rather than fully formal. Light notation is introduced to make the configurations concrete. The visual figures throughout serve as references that subsequent papers cite repeatedly. Heavy mathematical derivations and quantitative phenomena are distributed across the focused papers that develop specific topics: mass in Paper 8, gravity in Paper 10, time in Paper 13, beta decay in Paper 15, and particle composition in Paper 16. This paper provides the vocabulary; the subsequent papers apply it.

3. The Nine States: A Plain-Language Tour

Before any notation, here is the whole cast of characters. The framework says everything in the universe is one of nine kinds of geometric configuration — nine "states." Most of what conventional physics treats as a zoo of separate particles turns out to be these nine states, alone or bonded together. This section introduces all nine in plain terms: what each one is and why it matters. The machinery that makes them precise — the triangles, the three values, the tilt operations — comes in the sections after this, and will be easier to follow once the nine states are already in mind.

The single most important one is the photon, and the framework calls it **State 9**. It is the parent state: the flat, untilted configuration, the only one that travels at the full speed of light, and the baseline every other state is measured against. It is not "lower energy" than a massive state — energy is conserved throughout. What changes when a configuration departs from State 9 is not the amount of energy but its form: energy that was carried as pure forward propagation is converted into axis-tilt, and that tilt is what the framework calls mass. The configuration slows below c by exactly the amount its energy has shifted from propagation into tilt. Everything with mass is some departure of this kind from State 9, and is, in the long run, working its way back toward it. A few special behaviors belong to State 9 — it is the only massless state, the only one that runs no resolution search, and the state that collisions ultimately shed energy back into — and these recur throughout the paper.

The next, **State 1**, is the electron and the positron. It is the simplest configuration that carries mass: a fully resolved, self-contained loop that is stable on its own but sits below the speed of light because it has departed from the photon baseline. The electron and the positron are not two different substances; they are the same State 1 geometry tilted in opposite directions.

State 4 is worth flagging early because it has one special property, and an earlier version of this framework described that property wrongly. State 4 is *not* the most extreme or most distant configuration — it sits relatively close to the photon baseline. Its one distinction is what happens when three State 4 configurations bond at a shared center: the result is a perfect balance across all three axes, and that balanced moment is the unique point from which a heavy X-dominant configuration, an electron, and a photon can cleanly resolve. That is the only thing special about State 4; otherwise it behaves like an ordinary unbonded state.

The remaining states — **States 2, 3, 5, 6, 7, and 8** — are the "open" configurations. None of them is stable on its own; each carries unresolved geometry that drives it to bond with others in search of a stable, resolved structure. They are the building blocks that combine to form the heavy configurations (the structures conventional physics calls protons, neutrons, and the rest). What distinguishes them from one another is how much tilt each carries and in which direction, which the later sections make precise.

Two points about this cast. First, only two of the nine — State 9 (photon) and State 1 (electron-positron) — are stable standing alone; the other seven achieve stability only by bonding. Second, the heavy composites of ordinary matter are not new fundamental objects; they are these states bonded into resolved configurations, which is why the framework needs only nine states and no expanding particle zoo. With the cast introduced, the next sections build the geometry that defines them precisely.

4. The Three-Axis Interaction Fabric

The framework treats the universe as a geometric interaction fabric organized by three rotational axes, labeled Axis 1, Axis 2, and Axis 3. The three axes meet at a central junction point. Each axis is a rotational plane on which configurations carry integer values. The structure is not flat; it is three angular rotational planes intersecting at a common vertex, like three orbital planes meeting at one shared center.

Given the foundational three-axis, three-value geometric architecture, the 27 distinct configurations emerge as an inescapable structural consequence, completely bounding the state space. Each independent axis is completely devoid of arbitrary positive or negative signs, expressing instead exactly three unique structural integer states: X-axis tilt, Y-axis tilt, or the baseline letter O. This strict geometric constraint yields exactly $3^3 = 27$ fundamental structural configurations.

Within the framework, two features of the conventional particle spectrum are read as different views of the same underlying configuration rather than as separate fundamental entities. Matter and antimatter are the same off-axis geometric configuration distinguished only by tilt direction: an electron and a positron share the same structural departure from baseline, one expressing an X tilt and the other a Y tilt, rather than being two independent fundamental fields. And color charge is read as a localized rotational phase-shift of the coordinate fabric — the three "colors" being the three axes registering the same underlying tilt at different rotational positions — rather than three independent properties.

These are structural claims of the framework's own geometry, and they have an obvious consequence: the framework's twenty-seven configurations account for what the conventional spectrum catalogs as a substantially larger number of elementary states. Whether that larger conventional count reflects genuinely distinct fundamental species, or the same geometry counted several times over, is a question this paper raises but does not litigate here. The full case — including the history of how the factor of three entered physics as a patch, how collider apparatus produces the observed spectrum, and how the framework reads the R-ratio's factor of three as the three axes of the fabric rather than three color charges — is developed in Paper 2.5 (Section 7.1), with the quantitative reproduction reserved for the phenomenological treatment in Paper 16. Within Paper 2 the commitment is structural: matter-antimatter and color are geometric re-expressions within the twenty-seven-configuration space, not additional fundamental entities.

The interaction fabric is the medium in which configurations exist and through which they communicate. It is not a substance separate from the configurations; it is the geometric environment that the configurations themselves both shape and inhabit. When the framework speaks of gravity, expansion, or any other geometric phenomenon in subsequent papers, it refers to the configurations' relationships within this three-axis fabric.

4.1 The a_k Identifier and the Two Scans

The parameter a_k is not a measurement, a distance, or an active process. It is strictly the universal identifier — the structural punctuation mark that tells any discrete state, orientation, or configuration apart from anything else within the architecture. It does not scan, monitor, or register; it is the tag that establishes distinct identity,

marking where one configuration concludes and the next begins so that separate states do not overlap or blur on the system ledger. Assigning a spatial distance or metric length to a_k is a category error, and the framework relies on no quantum-gravity scale or Planck-length folklore to define it.

The actual mechanical work in the architecture is done by two scans, which are named here once and used by these names throughout the series:

- **The identity scan** — the inner 360° sweep by which a configuration confirms its own structural identity (whether it is the photon baseline, a resolved State 1, or an open state).
- **The temporal scan** — the outer 360° sweep by which a configuration carrying tilt searches its surrounding geometry for resolution. The accumulation of these outer sweeps is what registers the passage of time, which is why a tilted configuration experiences time and the zero-tilt photon does not.
- **The manifold scan** — the complete 720° cycle of both scans together, run by every configuration that carries mass.

The distinction between a_k and the temporal scan must be kept sharp, because conflating them is easy and wrong. The temporal scan is the active engine; a_k is the static label assigned to each distinct instance the scan produces. The scan does the work of moving the ledger forward; a_k simply marks each logged update as distinct from the last, which is what allows the ledger to record changes in a clear, forward-accumulating order. The one-way arrow of time is driven by the temporal scan's relentless execution, not by a_k — a_k is only the punctuation between the entries.

A clarification is needed regarding observational constraints on spacetime discreteness. High-energy astrophysical observations (such as photon arrival-time studies from gamma-ray bursts) place strong limits on certain models of discrete spacetime, specifically models in which discreteness produces an energy-dependent variation in the speed of light (Lorentz-invariance violation). The framework's discreteness is not of that kind. In this framework, the photon baseline propagates at c as the un-tilted parent state regardless of photon energy, because c is the baseline propagation speed of the zero-tilt configuration and does not depend on how the baseline is activated. The framework therefore does not predict energy-dependent photon dispersion, and the non-observation of such dispersion in high-energy astrophysical sources is consistent with the framework rather than a constraint against it.

This finite identification cycle is also what prevents the framework from inheriting the divergences that plague continuum field theories. The framework notes that mainstream mathematical paradoxes — specifically ultraviolet divergences and zero-point energy dependencies — are structural artifacts exclusive to continuum field theories and possess no meaning within this architecture. Because the photon is the absolute parent state, propagation at the capacity ceiling is the foundational baseline of the fabric. When energy is conserved into mass, the configuration is never at rest; it runs the continuous 720° manifold scan (the identity scan paired with the temporal scan). Because mass is a scanning cycle executing across a discrete coordinate manifold, the concept of a zero-distance point-particle is eliminated, rendering mainstream divergence crises obsolete. There is no "approach to zero distance" in this framework, so there is no infinity at zero distance to begin with; a_k is the punctuation that marks each interaction distinct from the next.

4.2 The Geometric Baseline and the Fallacy of the Absolute Vacuum

The background manifold is geometry. It is not a passive, un-activated container, nor a void awaiting occupation; it is the fundamental coordinate system from which all nine configurations emerge. A "photon" is not a separate substance laid over that background — it is simply a configuration in the zero-tilt baseline state, propagating at c . The distinction between "space" and "particle" is therefore not one of category but of state: the geometry is the substance, and what differs is only how a region is tilted.

Crucially, an "empty" vacuum chamber is a physical and structural impossibility. While mechanical pumps can evacuate localized, transient particle configurations from a chamber, the environment can never be truly empty. The containment walls of the apparatus, the surrounding magnetic shielding, and the laboratory housing are themselves dense, highly concentrated configurations of structural geometry. Because space is a continuous, unbroken three-axis matrix, the intense geometry of these physical boundaries continuously projects into, bleeds through, and defines the internal space of the chamber. There is no such thing as an absolute vacuum; there is only a continuous geometric field bound and shaped by the architecture of its containment walls. The photon (State 9) is not space itself; it is a fully resolved, zero-tilt excitation moving through this bounded geometric landscape.

5. The Three Discrete Values per Axis

The Axis-Tilt Operation: a discrete "click" off the baseline

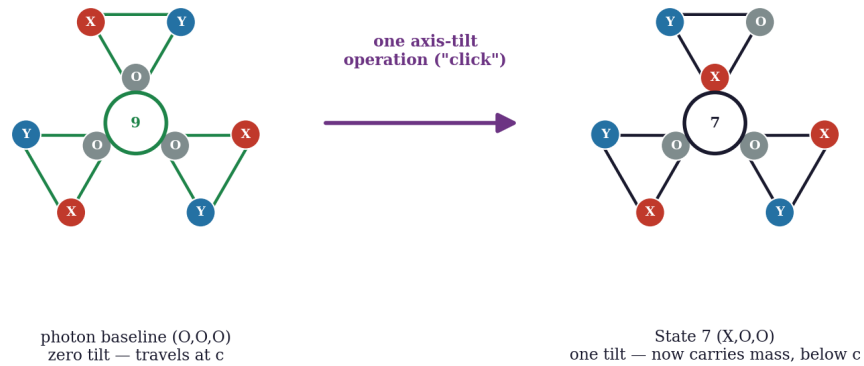


Figure 1: The axis-tilt operation: a single discrete “click moves a configuration off the photon baseline, converting propagation energy into tilt (mass).

Each of the three axes carries one of three discrete values at any geometric position. Throughout this paper and subsequent papers in the series, these values are denoted by the letters X, Y, and O.

The framework uses X, Y, and O as fundamental structural designators for a configuration’s orientation: each identifies the tilt state of a configuration along one of the three geometric axes relative to the photon baseline. They are discrete, categorical labels for geometric state — not numbers for arithmetic. (A reader who prefers signed notation may read X as a tilt one way, Y as the opposite tilt, and O as the zero-tilt anchor, but the letters are used precisely to keep these out of ordinary addition and subtraction.)

The labels X and Y are conventional and symmetric: tilt one way is X, tilt the other Y, and exchanging X with Y throughout produces the geometric mirror of a configuration. This symmetry is the structural basis for the framework’s treatment of antimatter (Paper 15) as the opposite tilt direction of matter rather than a different category of substance.

O serves as the framework’s structural anchor and identifier, preserving a configuration’s geometric identity; this is why State 9 — the photon baseline — requires three central O values to hold its stable, zero-tilt state. The active tilt values (X and Y) carry the structural energy and tension; the O anchors hold spatial reference steady, which becomes essential in bonding (Section 10), where O values at shared positions let geometric points be occupied without contradiction.

The choice of three discrete values per axis is adopted rather than derived. The framework does not allow continuous tilt values along an axis; tilt is quantized into these three states. This quantization is what makes the configurational structure enumerable. A continuous tilt parameter would produce uncountably many configurations and the framework's organizing power would dissolve.

6. Why the Photon Is the Parent State

The framework treats the photon (State 9) as the parent state — not as a metaphysical claim, but as a consequence of the geometry. The photon has zero tilt: its central junction is (O, O, O), its outer corners balanced. Running only its identity scan, its entire capacity budget goes into forward propagation, so it travels at c . Every other configuration is reached from the photon by accumulating tilt, and each can in principle return by undoing it.

6.1 The Tilt Cost, and the Velocity Ceiling

Mass, as defined in the Terms section, is one phenomenon at three levels: the geometric departure of a configuration from the photon baseline (what it is), the accumulated axis-tilt in its nine-position ledger (how it is measured), and the temporal scan that tilt activates (how it manifests). The mechanism connecting them is conservation of a fixed capacity budget whose ceiling is c . Energy is never lost, only redistributed: when axis-tilt operations move values between the central junction and the outer corners, energy that was carried as forward propagation is converted into tilt. Any tilt forces the configuration to run the temporal scan in addition to its identity scan, and the capacity diverted to that second loop is exactly what drags its velocity below c . More accumulated tilt means more mass and a velocity farther below c ; unwinding tilt brings the configuration back toward c .

This makes c a derived ceiling rather than an imposed constant. Nothing can run fewer scans than the photon's single loop, so nothing exceeds c ; and since mass is tilt, mass cannot drop below zero. The ceiling and the floor are the same fact stated two ways.

6.2 The Parent State and Resolution-Seeking

The photon is the parent state because it is the unique zero-tilt, zero-mass, c -propagating configuration; every other configuration is a perturbation carrying some accumulated tilt. That unresolved tilt is not inert — it produces real geometric tension, and that tension is what drives a configuration to bond or reconcile toward resolution.

A word on the "seeking" language used throughout this paper: it is shorthand for a mechanical process, not intention. An open configuration has unmatched values at its outer corners, and on a conserved manifold that imbalance creates a real restoring tension, exactly as a stretched rubber band or a displaced spring carries one. The configuration does not choose a partner; the tension mechanically drives it into alignment that relieves the stress, dropping it toward the lowest-tension state, the flat photon baseline. Wherever this paper says a configuration "seeks resolution," that is the physical meaning.

6.3 The Kinematic Gap

Two boundary conditions follow. A mass configuration cannot reach exactly $v = c$, because that would require undoing all its tilt at once, and unwinding proceeds only through actual bonding and unbonding events at the framework's discrete steps — so it approaches c asymptotically but cannot reach it without ceasing to carry mass. And it cannot rest at exactly $v = 0$, because carrying tilt means carrying unresolved tension, and unresolved tension is always driving rearrangement; even resolved configurations retain outer-bond activity (Section 10). There is no perfectly silent geometric state below c .

6.4 On the "Neutrino": A Collision Product, Not a Travelling Particle

The framework does not contain a neutrino. It is worth stating plainly why, because the neutrino is treated as settled fact, yet no experiment has ever observed one travelling: every "neutrino" result on record is the product of a collision inside a detector, not the sighting of a particle in flight. The entity was originally theorized purely as a patch — a ledger-balancing device proposed to account for energy that appeared to go missing in beta decay — and the framework regards it as exactly that. Positing an undetectable travelling particle to carry away missing energy makes no sense in this geometry and creates a conservation problem rather than solving one.

The framework's reading is mechanical and follows from conservation. When a collision delivers energy that does not resolve cleanly to a photon (the zero-tilt baseline) or to a resolved configuration, that energy cannot vanish and is not carried off by a ghost. It is conserved as new geometry produced at the moment of the hit. Crucially, the geometry that comes out is determined by what did the hitting: a photon impact resolves back toward a photon, while a State 5 impact produces State 5 geometry, and so on. This is precisely why the experiments "do not know what they saw" — the product is a configuration manufactured by the specific collision, and a configuration such as a State 5 is not visible as a photon and does not carry the electric-tilt signature of an electron-positron pair, so it leaves only an indirect trace. There is no travelling neutrino to detect; there is only the geometry each particular collision creates. The full deconstruction of the detection experiments — why a recorded collision product was mistaken for a pre-existing travelling particle — is developed in Paper 2.5.

7. The Three-Triangle Geometric Structure

A configuration in the framework is built from three triangles meeting at a central junction. Each triangle has three points: one at the central junction (shared with the other two triangles) and two at its outer corners. Each of these points holds a single integer value drawn from $\{X, O, Y\}$.

A complete configuration occupies nine positions in total. Three positions sit at the central junction — one center point per triangle, all meeting at the same geometric location. Six positions sit at the outer corners — two outer corners per triangle, distributed across the three triangles. Across all nine positions, the total content of any configuration is always the same: three X values, three Y values, and three O values. What differs between configurations is the distribution of these nine values across the central junction positions and the outer corner positions.

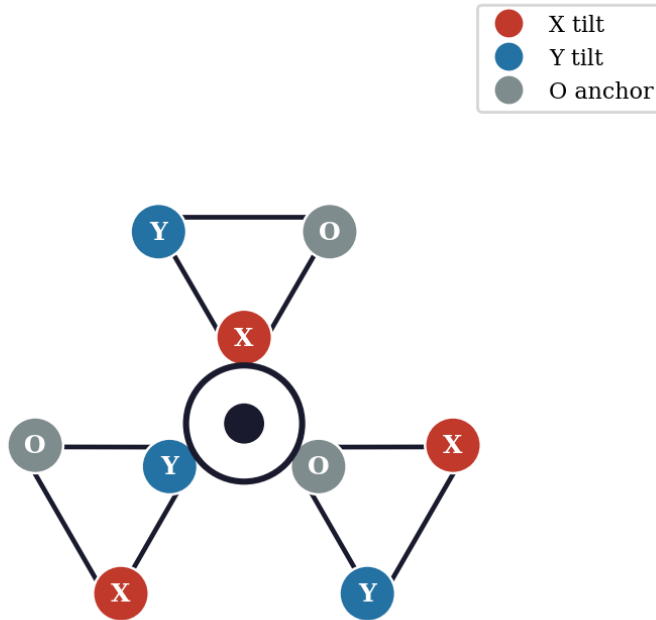
An ordered triplet used to label a configuration specifies the values at the central junction, listing the value on each of the three axes in order. The outer corner content is then determined by what remains. When the central junction carries one active axis-tilt and two O-anchors, it defines the configuration's unique structural identity, and the outer corners hold the balance of the configuration's fixed content (three X, three Y, three O across all nine positions). This is how the framework's accounting stays conserved: the total geometric content per configuration never changes, and the configuration's identity is set entirely by which values occupy the central junction.

Figure 1 shows this structure for the photon (State 9). The three triangles — T1, T2, T3 — meet at a central junction. Each triangle holds an O at its center point, a X at one outer corner, and a Y at the other outer corner. The three center points all sit at the central junction, giving (O, O, O) at center. Each triangle independently balances its X and Y outer corners. The whole configuration is resolved at baseline — no tilt anywhere, balanced everywhere — which is why the photon propagates at c .

7.1 The Axis-Tilt Operation Defined

An axis-tilt operation is the framework's mechanism for moving values between the central junction and the outer corners of a triangle. Specifically, an axis-tilt of one triangle is a rotation of that triangle about its radial

A Single Configuration (the radiation-symbol form)



each triangle carries three values: one at the center, two at the outer corners.
 The three center values name the state; every triangle still holds one X, one Y, one O.
 (total across the configuration: three X, three Y, three O)

Figure 2: A single configuration: three triangles meeting at a central junction, nine positions, content always three X, three Y, three O.

axis (the line from the central junction outward through the triangle's outer edge midpoint) such that one of its outer corner values rotates into the central junction position while the value previously at the center rotates out to an outer corner.

An axis-tilt is therefore a *discrete operation* that swaps a center value with an outer corner value within one triangle, while leaving the other two triangles' positions untouched. The total content of the configuration is conserved by an axis-tilt — no values are created or destroyed, only repositioned. What changes is the configuration's identity (which state category it belongs to, based on its new central junction content) and its accumulated tilt cost (mass).

For example, starting from a photon configuration where each triangle holds (O center, X outer, Y outer), an axis-tilt of one triangle that brings the X into the center produces a triangle holding (X center, O outer, Y outer). The configuration is no longer a photon; one axis has been tilted toward X-saturation. The configuration's central junction is now (X, O, O), which places it in State 7. The configuration carries mass corresponding to one accumulated tilt.

Subsequent axis-tilts can further reposition values across the three triangles. Three accumulated tilts in the same direction can produce a State 1 configuration (electron or positron). Mixed-direction tilts can produce intermediate state categories. Each tilt adds to the accumulated mass; each reverse tilt subtracts from it.

When two configurations interact and one transforms into another, what changes is the distribution of values across the triangle positions, achieved through one or more axis-tilt operations. The total content remains conserved across the interaction. This is the framework's analog to the standard conservation principles of physics, expressed in the framework's own language.

8. The $3 \times 3 \times 3$ Enumeration

The three axes, each carrying one of three values at the central junction, produce exactly $3 \times 3 \times 3 = 27$ possible center configurations. This is the complete set of states available in the framework. Every observable particle, every interaction product, every configurational state in nature corresponds to one of these twenty-seven configurations, or to a compositional structure built from combinations of them.

The twenty-seven configurations can be sorted by tilt distance — the number of axes carrying a nonzero value at the central junction, a measure of how far the center has departed from the photon's all-zero baseline. The distribution is one configuration at tilt distance 0 (the photon), six at distance 1, twelve at distance 2, and eight at distance 3, totaling $1 + 6 + 12 + 8 = 27$. Tilt distance alone is not the framework's most useful grouping, however; the finer classification by central-junction content signature (Section 8) partitions the twenty-seven into the nine working categories.

8.1 The Scaling Law and Tri-Configuration Convergence

The framework operates on a strict recursive scaling law: every structural advancement, from the photon baseline to complex bonding, is driven by the necessity of three-part convergence. The number three is the fundamental multiplier of the manifold, and it recurs at every level.

The system originates at the photon baseline (State 9), defined by three central O-anchors — the zero-tilt configuration in total structural equilibrium. To initiate the first mass-bearing state, the system requires exactly three axis-tilts to disrupt that equilibrium, establishing the minimal structural departure (State 1). Bonding among the open states (2 through 8) is then achieved at the outer perimeter: configurations align via matching integers at their outer vertices — an outside X meeting another outside X, a Y meeting a Y — and it takes three such configurations aligning their outer integers to finalize the bond, scaling the network outward.

State 4 occupies a unique position in this hierarchy. Its central junction can only bond with three total State 4 central junctions; when three State 4 configurations converge at their centers, they lock into a special

bond, and on completion the geometric tension is released as three outputs — a photon, an electron, and a positron.

The scaling law manifests when external geometric points intersect with internal structural points and a mismatch — an unresolved integer — is generated, the points failing to achieve a one-to-one mirror. Because the network cannot sustain an unresolved integer, it forces the recruitment of three additional configurations of the same state to complete the loop. This produces a six-point convergence cycle, which instantly resolves into two balanced three-point assemblies. Whether it is the three central O-anchors of the photon, the three tilts of State 1, the three configurations required for outer-vertex bonding, the three State 4 central junctions, or the three-part recruitment that resolves an unresolved integer, the system always returns to the base resolution of three — recruiting three more to bring the total to six, then dividing into two sets of three, preserving structural integrity without ever generating waste or breaking the conservation of geometry.

9. The Nine-State Classification by Content Signature

The Nine States (27 configurations by central-junction signature)

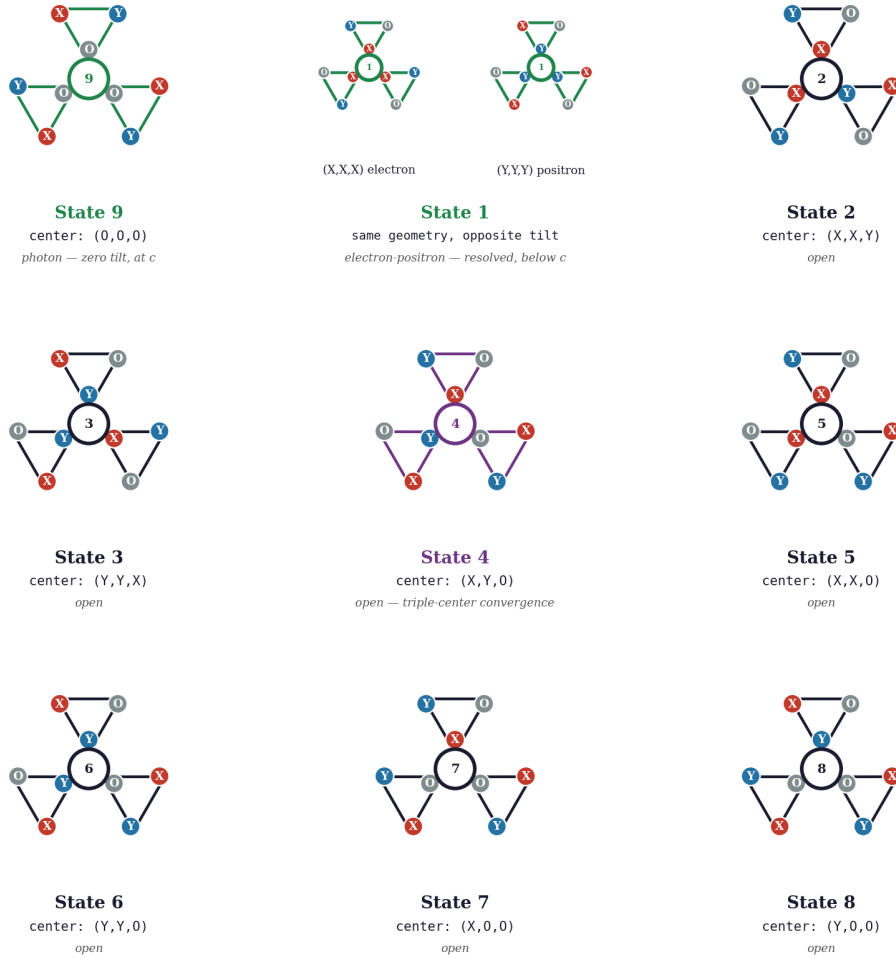


Figure 3: The nine states. Each of the twenty-seven configurations falls into one of nine categories by central-junction signature. State 9 (photon) and State 1 (electron-positron) are self-contained and stable; the rest are open.

Section 2 introduced the nine states in plain terms; this section gives their formal definition. The classification groups the twenty-seven configurations by the content of their central junctions: configurations whose centers share the same combination of values, regardless of which axis holds which, belong to the same category. The nine categories and their central-junction signatures are:

- **State 9** — (O, O, O), one configuration. The photon baseline, resolved at c .
- **State 1** — (X, X, X) and (Y, Y, Y), two configurations. The electron-positron pair, resolved but below c through three tilts.
- **State 2** — two X and one Y at center: (X, X, Y) and its orderings, three configurations.
- **State 3** — two Y and one X at center: (Y, Y, X) and its orderings, three configurations.
- **State 4** — one X, one Y, one O at center, six orderings. Relatively close to the photon baseline; its one distinctive property (the three-State-4 center convergence that resolves into a photon, electron, and positron) is established in Section 7.1.
- **State 5** — two X and one O at center: (X, X, O) and its orderings, three configurations.
- **State 6** — two Y and one O at center: (Y, Y, O) and its orderings, three configurations.
- **State 7** — one X and two O at center: (X, O, O) and its orderings, three configurations.
- **State 8** — one Y and two O at center: (Y, O, O) and its orderings, three configurations.

The counts sum to $2 + 3 + 3 + 6 + 3 + 3 + 3 + 3 + 1 = 27$, complete and consistent with the $3 \times 3 \times 3$ combinatorics.

9.1 Configuration Identity: Category, State, and Configuration

To ensure structural precision and prevent any ambiguity about what a "state" refers to, the framework distinguishes between the central coordinate seed and its total geometric expression using a strict three-tiered taxonomy:

a. Category: The broadest classification, determined solely by the combination of value types occupying the central junction regardless of orientation (for example, the combination of one X and two O values). The nine categories are defined at this level.

b. State: The specific spatial permutation and orientation of those three values at the central junction. Counting every distinct ordering of the central-junction values gives the twenty-seven distinct core states.

c. Configuration: The complete geometric realization of the structure, including the specific distribution of the remaining values across the outer corners, their loop resolutions, and bonding interactions.

Thus a State defines the exact coordinate seed of the central junction, while a Configuration describes the total geometric shape built around that seed. When the paper discusses bonding, resolution, and outer-corner interactions, it is operating at the Configuration level; when it counts the twenty-seven core states or the nine categories, it is operating at the State and Category levels respectively. This hierarchy resolves the question of whether a category such as State 7 is one object or several: it is one Category, comprising three distinct States (the three orientations of its central-junction signature), each of which anchors a complete Configuration once its outer corners and bonds are specified.

9.2 Classification Theorem: Invariant Topographical Completeness

The partition of the twenty-seven configurations into nine categories is not an arbitrary grouping. It is the mathematically inevitable consequence of distributing three symbols across three positions, and it can be stated as a theorem.

Theorem 1 (Invariant Topographical Completeness). Let a discrete three-axis junction configuration be defined by an ordered triplet where each axis independently selects a structural parameter from the closed alphabet $\Sigma = \{X, Y, O\}$. The total state space consists of exactly $3^3 = 27$ unique discrete configurations. Under the action of the symmetric group S_3 (axis permutations), these 27 states partition uniquely and exhaustively into exactly nine invariant equivalence classes, distributed according to the multinomial expansion:

$$\sum_{i=1}^9 C_i = 2 + 3 + 3 + 6 + 3 + 3 + 3 + 3 + 1 = 27$$

- Class of symmetric saturation (count 2): $\{XXX\}$ and $\{YYY\}$ — the absolute outer-limit parameters (State 1).
- Classes of asymmetric duplication (count 3 each, six such classes): $\{XXY\}$, $\{XXO\}$, $\{YYX\}$, $\{YYO\}$, $\{OOX\}$, $\{OOY\}$ — the mass-tilt shifting configurations.
- Class of complete asymmetry (count 6): $\{XYO\}$ — the six orderings of one-of-each-value (State 4).
- Class of the null anchor (count 1): $\{OOO\}$ — the perfectly flat parent baseline (State 9).

Proof. The distribution is governed by the partitions of three positions populated by a three-symbol alphabet. A triplet drawn from a three-symbol alphabet falls into exactly one of three partition types by symbol multiplicity: all three identical (multiplicity 3), two identical and one distinct (multiplicity 2+1), or all three distinct (multiplicity 1+1+1). The all-identical type yields 3 orderings collapsing to 1 class each, and only two of the three symbols (X, Y) produce the saturated outer-limit while the third (O) produces the null anchor. The two-and-one type yields 3 distinct orderings per class. The all-distinct type yields $3! = 6$ orderings in a single class. Enumerating these across the alphabet produces exactly the nine classes above, summing to 27. Because the partitioning is exhaustive over all possible triplets, no twenty-eighth state can exist and no signature can be bypassed. The architecture is structurally complete and topographically exhaustive, completing the proof.

This theorem establishes that the twenty-seven configurations and their nine-category partition are not a list the framework chose but a closed combinatorial law the framework is bound by. Given the adopted three-axis, three-value architecture, the state space has no freedom: it must take exactly this form.

10. The Resolution Principle: Intrinsic and Compositional

The framework's resolution principle determines which configurations achieve stability. To prevent any ambiguity, the framework establishes a strict distinction between two kinds of resolution: the intrinsic resolution of a single configuration, and the compositional resolution of a multi-node network. This distinction is essential because the same word "resolved" applies to both State 1 and State 9 (single configurations that close on their own) and to resolved configurations (multi-state structures that close collectively). They are resolved in different senses, and conflating the two senses produces apparent contradictions. The full derivation and analysis is developed in Paper 5; what follows is the working statement needed to ground the bonding discussion in Section 10.

10.1 Two Kinds of Resolution

Intrinsic resolution (single-configuration resolution). State 9 (the photon baseline) and State 1 (the electron-positron saturated anchor) are the only single states that possess an intrinsic resolved state on their

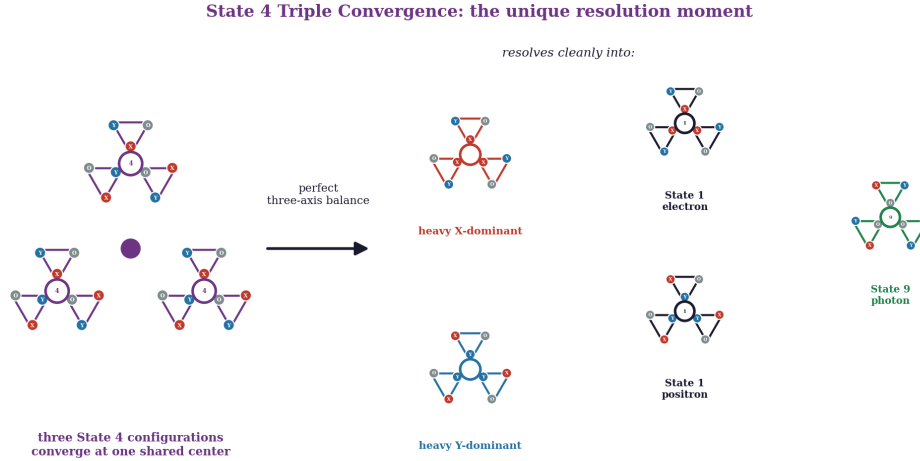


Figure 4: State 4 triple convergence: three State 4 configurations meeting at a shared center produce a perfect three-axis balance, the unique moment from which a heavy X-dominant configuration, a heavy Y-dominant configuration, an electron, a positron, and a photon cleanly resolve.

own. Their internal geometric loops are self-contained within their independent nine-position matrices. They exist stably without any external structural assistance. No bonding is required; the single configuration is complete.

Compositional resolution (multi-node collective resolution). Every other state category — States 2 through 8 — is inherently open as a single configuration. These categories cannot achieve stability alone. To form a stable resolved configuration, an open configuration must combine with one or more other configurations into a multi-node interpenetrating network. The network achieves resolution collectively: the unmatched integers across all the constituent configurations combine and balance, even though no individual constituent is resolved on its own. For example, three open configurations must combine their open ends into a three-node resolved configuration to collectively satisfy the resolution requirement.

The key point is that "resolved" means the same structural thing in both cases — no unresolved integer remains searching for a partner — but the scale differs. Intrinsic resolution resolves within one configuration's nine positions. Compositional resolution resolves across the combined positions of a multi-node network. When Section 10 describes resolved configurations as "resolved," it means compositionally resolved: the three-node network has resolved all its unmatched integers collectively, producing a stable composite even though its single-axis constituents (States 2 through 8) are individually open.

10.2 Why Only Two Arrangements Close Intrinsically

Intrinsic resolution is possible in exactly two ways, and the content rule (three X, three Y, three O across nine positions) is what limits it to two. To leave no outer corner carrying an unresolved integer, a configuration must concentrate either all three O values at the central junction — the photon (State 9), zero tilt, no mass, traveling at c — or one saturated tilt value at the junction, (X, X, X) or (Y, Y, Y) — the electron-positron pair (State 1), which carries real accumulated tilt and so has mass and travels below c , yet is geometrically resolved and therefore stable. Every intermediate distribution (one tilt with two O, two matching tilts and an O, or one of each) leaves at least one outer corner with an unmatched integer, which is exactly what makes States 2 through 8 open and dependent on compositional resolution.

The two intrinsically resolved states thus differ in velocity for a clear structural reason: State 9 closes at zero tilt and runs only its identity scan, so its whole capacity goes to propagation at c ; State 1 closes at full

tilt and must also run the temporal scan, and the capacity diverted there is what holds it below c. Two stable anchors, different mass and velocity, same resolution principle — with everything between them open.

11. Bonding and Chain Progression

First-Link Bond: the outer-to-center docking (dual-axis-plus-one-surplus)

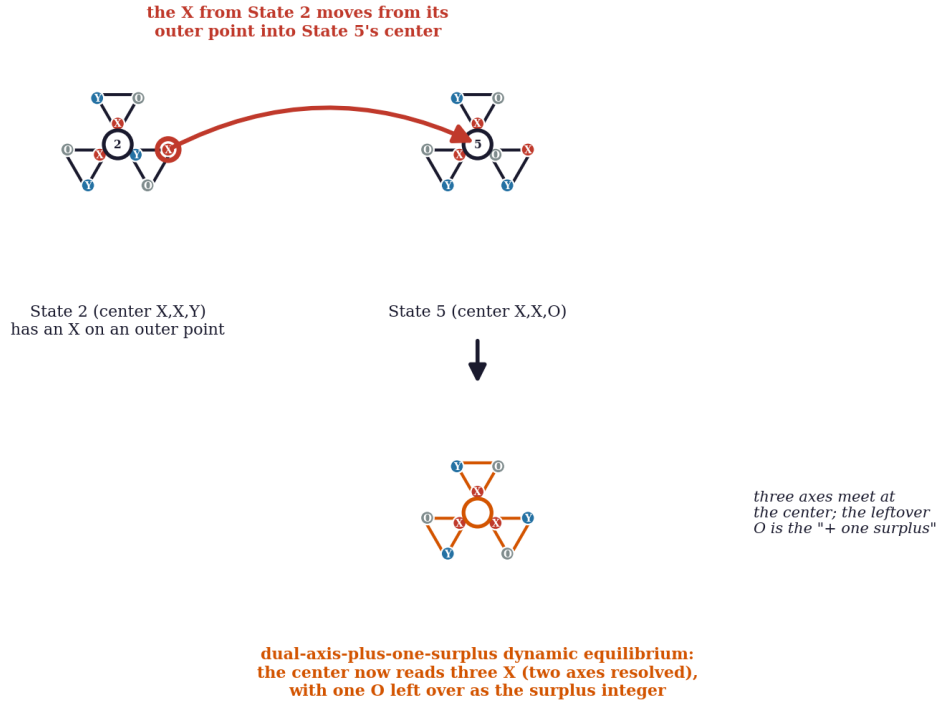


Figure 5: First-link bond (dual-axis-plus-one-surplus): the X on an outer corner of State 2 docks into the center of State 5, giving three X at the center (two axes resolved) with one O left over as the surplus integer.

The framework's bonding mechanism works through the outer corner positions of configurations meeting either the central junctions or the outer corners of other configurations. Specific rules govern which integers can meet which.

Before the mechanics, the framework names the structures that bonding produces, and it names them by their geometry rather than by borrowed particle nouns. The naming counts two things: how many tri-axis units have come together, and how much unresolved surplus sits at the junction. A **tri-axis dynamic equilibrium configuration** is a single self-contained unit in balance — State 1 and State 9, which stand alone. When the open states (2 through 8) meet at their outer accesses, each one brought to the junction adds its axes, so three of them converging is a **triple tri-axis dynamic equilibrium configuration** — the fully resolved configuration. Between standing alone and full resolution lie the surplus-carrying stages, where "surplus" means unresolved integers left at the junction. One outer corner meeting one inner junction leaves two axes resolved and one over, a **dual-axis-plus-one-surplus dynamic equilibrium configuration**. When that structure bonds further so that all three axes are resolved but one extra integer remains, it is a **tri-axis-plus-one-surplus** configuration; when two extra integers remain, a **tri-axis-plus-two-surplus** configuration, which may be X-dominant or Y-dominant depending on which tilt the surplus carries. A surplus is only a surplus while it is incomplete: three surplus integers that come together into a complete matching triplet are no longer surplus — they have resolved into resolution. This vocabulary replaces the

Triple Tri-Axis Dynamic Equilibrium: three matching outer corners converge

three configurations, each presenting a matching X on an outer corner

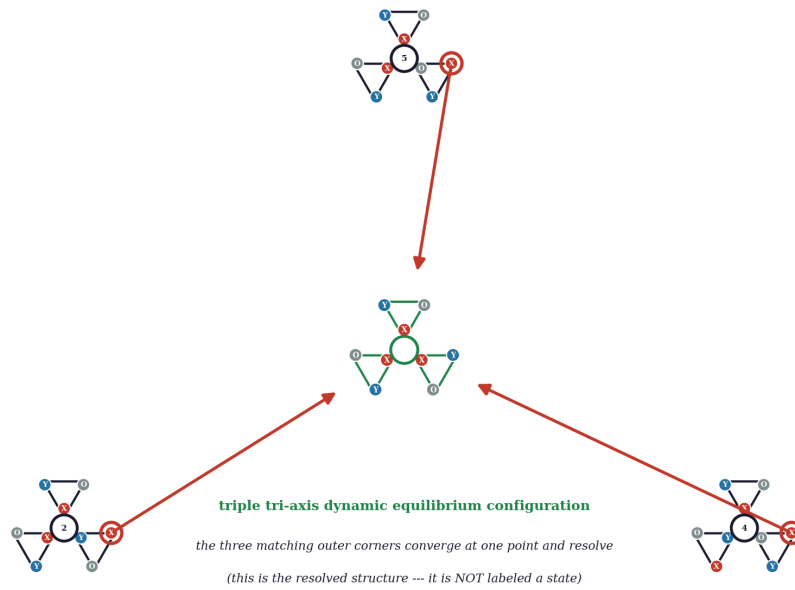


Figure 6: Triple tri-axis dynamic equilibrium: three configurations each present a matching outer-corner value; the three matching corners converge at one point and resolve into the triple tri-axis structure (the resolved configuration — not itself a state).

conventional meson/baryon/exotic naming entirely; where a conventional name aids the reader it is given once as a bridge.

11.1 The Single-Integer Matching Rule

The outer corner of a configuration holds a single integer (X, O, or Y). Two bonding modes are available.

Mode 1: Outer-to-center docking. An outer corner of one configuration docks into the central junction of another. The docking is valid when the receiving central junction has at least two matching integers already in place, AND the receiving configuration is not already resolved. For example, an outer corner holding Y can dock into a central junction that holds (Y, Y, O) or (Y, Y, X), because two Y values are already present at the junction and the docking Y contributes a matching third value to settle that center position. A center holding (Y, Y, Y) does not accept docking because it is already in the saturated resolved state of State 1 — the resolution principle is satisfied and no additional integer is sought. If the central junction already possesses these three values, an incoming Y will actually be repelled. This repulsion is not driven by positive or negative charge mechanics; rather, because the structural integer value is already completely resolved, the junction has no geometric capacity for further entry and actively repels the incoming value.

Configurations whose outer corners hold X can dock into open centers containing two or more X values (such as (X, X, O) or (X, X, Y)). Configurations with Y outer corners can dock into open centers containing two or more Y values (such as (Y, Y, O) or (Y, Y, X)). Configurations with O outer corners can dock into open centers containing two or more zeros (such as (O, O, X) or (O, O, Y)). Thus, you would get the (O, O, O) value in the center. While this establishes the central junction value of a photon, the overall configuration still carries axis-tilt and mass from its remaining structural positions, meaning it is explicitly not a photon. Instead, this mechanism defines the exact geometric pathway by which a photon-value junction is allowed to be brought back and to go away within a matter configuration.

Two categories naturally fall outside the docking pool because they are already resolved. State 9 (center (O,O,O)) and State 1 (center (X,X,X) or (Y,Y,Y)) have satisfied the resolution principle and have no unresolved integer at their centers to receive a docking partner; their geometric structure is settled. State 4 (center with one of each value: X, Y, O) is open but has no two-matching-integer slot at its center, so it cannot accept a single docking integer in the standard outer-to-center mode. Three State 4 configurations could in principle interpenetrate at a shared center if all three contribute the same single integer to that center, but this is a special case rather than a routine bonding path.

Mode 1 applied to State 4: State 4 can still bond actively through its outer corners. Its three triangles hold the values that were displaced from its central junction during whatever tilt history produced it; those outer corners carry single integers that can dock into other configurations' centers (Mode 1) or converge with matching outer corners of other configurations (Mode 2). State 4 is fully participatory in bonding chains; what is restricted is the docking direction at its own center, not its capacity to bond at all. The genuinely distinctive event occurs when three State 4 configurations combine at a mutual center point. Because each contributes one X, one Y, and one O, the convergence produces a perfect simultaneous balance across all three axes — the unique geometric moment, confirmed by the integer algebra, from which a heavy X-dominant configuration, an electron, and a photon can be cleanly resolved. This is the only special property the State 4 center-bond carries; it is not a maximal-tilt or maximally-strained intersection, and earlier descriptions of it as carrying an inescapable surplus far from the baseline were mistaken.

When this balanced three-axis junction is forced to resolve — which, as developed in Paper 2.5, occurs when an actual collision or an intrusive laboratory boundary geometry drives the resolution rather than as a spontaneous unpacking — the products are determined by the collision itself, not unpacked from inside. This is the central correction the framework makes to the conventional beta-decay story. Standard physics records fragments emerging from a disturbed nucleus and assumes they pre-existed within it. The framework holds instead that the energy delivered by the hit is conserved as new geometry, and the geometry that comes out

is whatever the impacting configuration was: a photon impact resolves toward a photon, a State 5 impact produces State 5 geometry, and so on. Nothing was hiding inside; the apparatus that drove the collision determined what was produced.

This is also why the conventional account reached for an undetectable particle. When the energy of a hit does not resolve cleanly to a visible photon or a recognizable electric-tilt configuration, the product is a geometry the detector cannot read directly — a State 5, for instance, is neither a photon nor an electron-positron signature — so a portion of the energy appears, on their instruments, to have gone missing. They posited the neutrino to balance that apparent deficit. The framework needs no such particle: the energy is fully conserved as the collision-produced geometry, whether or not that geometry happens to be one the detector is calibrated to see. The resolution is non-invertible — the collision injects an irreversible update into the local coordinates — which is what ties this event to the forward direction of the ledger.

Mode 2: Outer-to-outer convergence. Three matching outer corners from three different configurations converge at a single meeting point. Three outer corners each contributing the same integer (three X values, three O values, or three Y values) settle that meeting point as a three-integer position equivalent to a configuration's central junction. This is the cleanest bonding event, producing the lightest stable multi-state structure the framework recognizes. To be precise: the single self-contained states (State 9 photon and State 1 electron-positron) are stable on their own without bonding; the convergence bonding produces the lightest stable structure built from *multiple* configuration states, which is the lightest resolved configuration.

11.2 The Three Outcomes of the External Sweep

Particles do not blindly bond with anything nearby. During the temporal scan, a configuration performing the audit makes a strict mechanical assessment: will this neighbor help me achieve resolution, or will it make my imbalance worse? The sweep is a strict mechanical valuation based on baseline resolution, with three possible outcomes:

Attraction and bonding. If a neighboring configuration presents a complementary integer value that resolves the localized tension, a mutual structural pull is established. This geometric minimization of tension is observed macroscopically as attraction, culminating in a bonding operation.

Repulsion. If a neighboring configuration presents an incompatible integer value (for instance, already-resolved values, or values that would compound the existing imbalance), a bond is mechanically rejected because it fails to provide a resolution pathway. The configurations actively force one another away, a geometric defense mechanism observed macroscopically as repulsion.

No interaction (stays neutral). If a neighboring configuration offers neither a resolving match nor a conflicting obstruction, nothing happens between them — the configuration simply remains as it is. This is not isolation: nothing in the framework is ever isolated, because the integers are the fabric and every configuration is always surrounded by geometry. It is the neutral case, where the surrounding geometry neither completes the configuration's resolution nor compounds its tension, so no bonding and no repulsion occurs.

This screening rule unifies the three possible outcomes as a single mechanical process: the temporal scan valuates each neighbor against the configuration's drive to return toward the State 9 photon baseline, and either bonds (resolution available), repels (resolution blocked), or does neither and stays neutral (resolution neither available nor obstructed). The detailed treatment of repulsion as a destination-ledger failure rather than a sign-based rejection is developed in Section 12.

11.3 First Link: Dual-Axis-Plus-One-Surplus Configuration

A first link forms when an outer corner of one configuration docks into the central junction of another, or when an outer-to-outer convergence begins. The configurations are now bonded, with two axes resolved and

one integer left over at the junction — a dual-axis-plus-one-surplus configuration. That single surplus integer needs resolution.

For example, State 8 has one of its three outer corners holding Y. When that Y outer corner docks into another configuration's central junction that already holds (O, Y, Y), the docking Y fills the matching slot. But the receiver had an O at its central junction in addition to the two Y values, and that O now exists alongside the new triplet of Y values in a way that leaves the configuration's overall geometry with at least one outer corner still searching for resolution. The first link is unstable.

The bond is real, and the configurations stay together long enough for detectors to register a signature — which is why first-link states are observable in collision experiments. But the structure has not reached resolution, and if it cannot develop toward resolution it breaks down: a link that offers no path back toward the photon state releases, and the configurations resume their individual paths. The stages that follow are named by their geometry, as the next subsections set out.

11.4 Tri-Axis-Plus-One-Surplus Configuration (Second Link)

A second bonding event develops the structure further. When it leaves all three axes engaged but one integer still unresolved at the junction, the result is a tri-axis-plus-one-surplus configuration — more developed than the first link, still carrying surplus, still searching for resolution. These surplus-carrying configurations are the transient force-carrier-like states standard physics observes in collisions and decay channels (conventionally, mesons); the "force-carrier" behavior is simply their active search for the bonding event that would resolve the surplus into a resolved triplet. Two of them may bond briefly and gain apparent stability, but the combined surplus typically stays unresolved, so the temporary bond breaks and each resumes its search. Lacking the resolving value, a surplus-carrying configuration that cannot complete its junction inevitably decays.

11.5 Triple Tri-Axis Configuration (Resolution Achieved)

A triple tri-axis configuration — the fully resolved structure — forms when the chain accumulates enough bonding events to resolve all surplus integers, so that every outer corner has either docked into a matching center or converged with two others at a single meeting point. The lightest form through the cleanest path: three configurations whose outer corners all hold the same value converge at one point, settling three matching integers with no surplus. Heavier ones form through more complex paths — multiple outer-to-center dockings accumulating surplus across several bonding events before resolution is reached.

This gives the mass hierarchy a structural cause. Each bonding event accumulates tilt-cost; a triple tri-axis configuration reached through one clean convergence carries minimal accumulated tilt, while one reached through multiple dockings carries more. More accumulated tilt is more departure from the photon state, hence more mass. The framework does not yet derive absolute mass values — the structural ordering (light from simple convergence, heavy from accumulated dockings) is established here, and the specific values across the spectrum are the work of Paper 8.

11.6 The Heavy Configurations Defined Structurally

The framework identifies the observed heavy composites with specific multi-state network structures directly from the twenty-seven-configuration manifold, and it names them by their geometry rather than by the conventional particle names, because those names carry the assumption of pre-packaged internal contents that the framework rejects.

The naming is worth stating plainly before it is used, so it does not read as jargon. "Heavy" means a resolved three-node composite carrying substantial accumulated tilt — a resolved-configuration structure rather than a single configuration. "Dominant" names which way that structure's net tilt leans. A **heavy X-dominant configuration** and a **heavy Y-dominant configuration** are the *same structure* differing only

in the direction of their tilt — the same way the electron and positron are one geometry spun in opposite directions, with neither direction privileged over the other. There is no "positive" or "negative" here, only which axis the tilt favors. What the conventional account calls the proton and antiproton are, in this reading, simply these two tilt directions of one structure.

The genuinely distinct case is what the conventional account calls the neutron. It is not a third tilt direction of the same composite; it is a different geometry entirely — three State 4 configurations merging at a single mutual center point. That three-way center merger, not a tilt direction, is what sets it apart, and the framework names it the **heavy balanced configuration** because its tilt is globally balanced rather than axis-dominant. The geometric names are used as primary throughout the series, with the conventional name given on first use as a bridge and restated later only where a reader would otherwise lose the thread.

The heavy X-dominant configuration (conventionally, the proton) is a three-node network of distinct open configuration states whose outer corners meet at a convergence point. Combining three distinct configurations from the two-axis-tilt-and-an-O categories (State 5, two X-axis tilts and an O, with State 6, two Y-axis tilts and an O) lets their outer corners converge so that the X-axis tilts, the Y-axis tilts, and the O values each balance across the network into matched (X,X,X), (Y,Y,Y), and (O,O,O) groupings. Because it takes three distinct configurations to close that network, it is a complex resolved configuration with a stable, fully resolved net tilt. Its axis-dominance follows from the tilt direction it inherited from the initial operations, manifesting as either an X-dominant or a Y-dominant structure across the surviving outer bonds. (The opposite tilt direction is the **heavy Y-dominant configuration**.)

The heavy balanced configuration (conventionally, the neutron) is not a composite with smaller particles packed inside it. It is a singular configuration formed by three State 4 configurations converging at a mutual center point. Because each State 4 contributes one X, one Y, and one O, that convergence produces a perfect simultaneous balance across all three axes — the same balanced triad moment described in Section 10, here standing as a single globally balanced structure rather than resolving apart. This is the central structural commitment: there is no matter hiding inside matter. It does not contain a heavy X-dominant configuration, a State 1 configuration, and a residue waiting to be released; it is one balanced configuration whose stability or instability is a matter of how its three-axis balance is held or disturbed, not of pre-packaged parts straining to escape.

This raises the obvious question: what, then, are the products observed in beta decay? The framework's answer is that they are configurations resolved into existence at the moment of measurement, when the apparatus imposes its boundary geometry on the high-tension junction — not parts that pre-existed inside. The full experimental case (the beam, bottle, and energy-spectrum experiments) is the work of Paper 2.5. Within Paper 2 the commitment is only structural: when forced to resolve, the balanced configuration casts off a fragment of its perimeter tilt as a standalone State 1 configuration, leaves a core remnant that registers heavy inertial resistance under an accelerating field, and produces a transient strain in the surrounding fabric — and none of these pre-existed as packaged parts.

These identifications map the framework's configurations onto the observed heavy composites through specific, checkable structural assignments, while rejecting the assumption that observed decay products were pre-existing internal parts. The heavy X-dominant configuration as a distinct-state convergence and the heavy balanced configuration as a triple-State-4 junction are worked examples; the full spectrum of heavy composites (conventionally the Delta, Sigma, Omega, and excited states) is developed in Paper 16.

This external resolution-seeking is the structural origin of two phenomena developed in subsequent papers. The fabric's accounting of the inward consolidation paths that the resolved configuration's external geometry seeks is what the framework identifies as gravity (Paper 10). The outward search of unresolved open geometry for resolution is what the framework identifies as the driver of cosmic expansion (Paper 14).

It is important to be precise about which configurations drive expansion. Expansion is driven by the open states (States 2 through 8) whose outer values are searching for resolution. These open states carry unresolved outer geometry, and conservation requires that the unresolved values must eventually find resolution. That

conserved outward search — open values reaching outward for the partners that would complete their resolution — is the mechanical cause of expansion. The self-contained systems do not drive expansion: State 9 (the photon) is resolved at the baseline and seeks nothing, and State 1 (the electron-positron) is internally resolved and complete. Resolved composite structures bound inside galaxies and planets are likewise internally resolved and do not contribute to the outward expansion search. Expansion is a property of the open, unresolved territory, not of the resolved matter resting within it.

This also explains why the expansion-driving geometry is not directly visible. The open states searching for resolution are not bound into light-emitting composite matter; they carry unresolved geometry across the baseline coordinate manifold without producing visible light, so the expansion-driving territory reads as darkness. In a galaxy or a planet, by contrast, the configurations are tied together as heavy and resolved composites — stable and visible. The visible universe is that resolved-configuration matter; the dark, expansion-driving territory is the open states conducting their conserved outward resolution search.

The framework recognizes that connecting microscopic open-state dynamics to megaparsec-scale cosmic expansion requires an integration mechanism, and outlines that mechanism here, with the full quantitative development deferred to Paper 14. Three structural rules bridge the scales. First, expansion is spatially segmented: galaxies, stars, and planets are dense clusters of resolved configurations whose local coordinate systems are already locked and anchored, so geometric expansion cannot occur inside them; cosmic voids, by contrast, are wide-open territories of unresolved open states that act as the pressure-valves absorbing the outward displacement of the fabric. Second, there is a scale delay: a single microscopic resolution event produces a displacement far too small to measure at the moment of occurrence, and the metric displacement only becomes telescope-measurable after millions of separate multi-state resolutions have aggregated over cosmic time. Third, the mechanism predicts epoch-dependent acceleration: as the universe evolves and generates more resolved-configuration matter, the frequency of multi-state corner-convergences rises, each resolution forces a corresponding outward displacement into the unbroken network, and so later epochs carry a higher density of unresolved outward search than earlier ones. The framework reads this rising displacement density as the mechanical cause of what standard cosmology measures and attributes to an accelerating dark-energy-driven expansion. The claim here is structural: the same geometric resolution process operates at both the local scale and the cosmic scale, with the integration mechanism above bridging them. The quantitative match to the observed Hubble parameter and the acceleration history is the work of Paper 14.

Gravity and expansion thus share a common structural origin in resolution-seeking, but they are driven by different participants: gravity by the inward consolidation that resolved configuration matter induces in the surrounding fabric, and expansion by the outward resolution-search of the open, unresolved states. The two phenomena are the inward and outward faces of the same geometric drive toward resolution.

11.7 Tri-Axis-Plus-Two-Surplus Configuration: The Resolution Search Exhausts

A tri-axis-plus-two-surplus configuration (what the conventional account calls the tetraquark, and the heavier exotic states as well) is a structure that briefly forms during a collision and then decays because it carries two surplus integers at the junction — more than a clean resolution can accommodate. It may be X-dominant or Y-dominant depending on which tilt the surplus carries. The framework's account of why these are observed but transient is as follows.

When four configurations attempt to bond into a single chain, the accumulated leftover integers across the four exceed what cubic resolution can accommodate. The structure begins to form because the individual bonding events are geometrically valid — each outer-to-center docking or outer-to-outer convergence happens normally. But the accumulated chain cannot resolve to a stable resolved configuration. The internal geometry searches for a path to resolution; the search runs through the available configurational arrangements; no path resolves.

This search-and-fail process is what makes the over-constrained configuration observable but transient. It exists during the search phase — long enough for detectors to register a signature. When the internal search exhausts all possibilities and finds no resolution, the structure reverts: the excess is shed, and what remains collapses to a resolved configuration plus shed open remnants, or to constituent configurations dispersing toward the photon state.

Multiple paths produce these formations: two open configurations searching for resolution can bond temporarily; a resolved configuration with lingering outer geometry can accept an extra configuration at its open positions; or an over-bonded chain can attempt to form in a high-energy collision before the resolution search confirms whether resolution is achievable. In each case it forms because the local bonding events are valid, persists while the search runs, and decays when the search exhausts.

The LHCb collaboration has observed several such states (conventionally, tetraquarks) in collision data. The framework reads these as transient over-constrained structures that formed because the local bonding was geometrically valid and decayed when no resolution path was found. The detailed account is developed in Paper 16 and in Paper 22 of the critique tier.

11.8 What This Establishes

The framework's account of stable matter, transient open configurations, stable resolved configurations, and unstable over-constrained configurations reduces to one principle: resolution of the geometric structure through accumulated bonding events. Configurations that close on their own (State 9 photon, State 1 electron-positron) are stable. Open configurations are transient unless they bond into composite structures that achieve resolution. The progression dual-axis-plus-one-surplus $\rightarrow$ tri-axis-plus-one-surplus $\rightarrow$ triple tri-axis corresponds to accumulated bonding events resolving surplus integers until resolution. The tri-axis-plus-two-surplus category corresponds to attempts that exceed what resolution can accommodate.

Resolved configurations close internally but their outer bonds continue searching for resolution toward the photon state. This outer search is what registers as gravity and contributes to cosmic expansion in subsequent papers.

12. Interaction, Force, Energy, Momentum, Time: Brief Statements

This section provides brief statements of how the framework treats several physical quantities that subsequent papers develop fully. These statements are not derivations but structural placeholders that orient the reader to where each topic lives in the framework's geometry.

12.1 Interaction

An interaction occurs the moment the outer position of one configuration shares a coordinate with the central junction of another. It does not matter whether the shared position holds an X, Y, or O value—the trigger is strictly geometric proximity and axis alignment, not contact across empty space, because configurations are the fabric itself.

Each such event is marked distinct by a_k (the identifier defined in the Terms section). This removes any need for force carriers across empty space: there is no empty space in the interaction fabric, every point of which is occupied by some geometric state. Interaction is direct geometric alignment between adjacent configurations, not action at a distance mediated by separate particles.

The ledger-interface rule. A natural question arises the moment two configurations touch: if each configuration carries an invariant ledger of three X, three Y, and three O values, does that conservation break down during an interaction? It does not, and the rule is strict. During any localized collision or multi-node merger, both participating configurations preserve their individual nine-position identity ledgers. Neither structure yields, alters, or exchanges its native allocation of three X, three Y, and three O values.

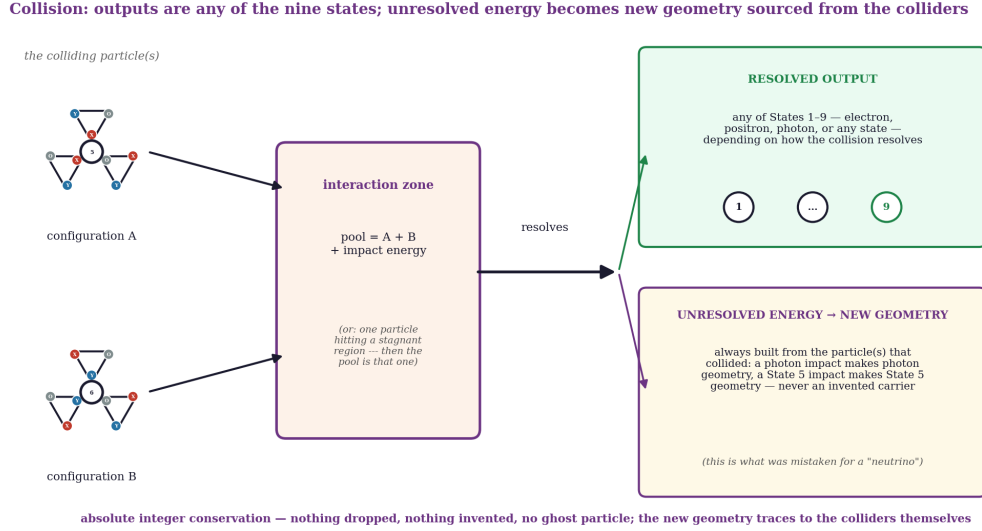


Figure 7: A collision conserves energy as new geometry. The combined pool of both configurations plus the impact resolves into full nine-position states — no energy lost, no ghost particle required.

The interaction is driven entirely by the distribution of the external impact energy, which forces independent axis-tilt operations across both networks to reallocate the added kinetic geometry, leaving the core identities of the parent configurations structurally invariant throughout the cycle.

The collision addition rule. A collision is governed strictly by state addition and redistribution. An interaction requires an incoming configuration and a target configuration, each maintaining its individual nine-position invariant ledger. The high-impact collision introduces extra kinetic geometry into the localized interaction zone, so the total structural pool of the event is the sum of both parent ledgers plus the added impact geometry. That energy influx forces sequential axis-tilt operations across the triangles, driving the configurations toward new designations — a state driven toward the photon baseline, say, or a multi-node merger. When the interaction zone resolves, the combined pool cleanly divides back into fully compliant, standalone nine-position states, ensuring absolute integer conservation from inception to resolution. When a collision produces new geometry, that geometry is itself a compliant nine-position state, and the integers composing it come entirely from the combined pool of the event; nothing is dropped, nothing is invented, and no undetectable carrier particle is required to balance the books.

This mechanism is directly visible in the production of photons in high-energy collisions. Under extreme impact, the added collision geometry forces a configuration's axis-tilts to unwind until its central junction reaches the (O, O, O) baseline; at that threshold the configuration is fully resolved, sheds all mass-drag, and propagates natively at c as a photon. The process is reversible: the resolved state remains stable until a subsequent fabric intersection introduces new geometric energy, re-engaging the axis-tilts and returning the ledger to a massive designation. Conservation holds in both directions, because the integer pool is only ever redistributed, never created or destroyed.

12.2 Forces

The framework eliminates force carriers as separate entities and replaces them with the resolution-seeking mechanism that drives bonding. What standard physics treats as four fundamental forces — gravity, electromagnetism, weak, strong — are all manifestations of configurations seeking resolution toward the photon state through their available geometric paths.

Gravity is the fabric's accounting of resolved configurations' outer-bond resolution-seeking (developed in Paper 10). Electromagnetism is the framework's account of charge as axial unwinding direction (Section 12). The weak interaction is the framework's account of beta decay through axis-tilt operations that transform one configuration into another (Paper 15). The strong interaction is the framework's account of bonding within resolved configurations that holds them together (Paper 16).

None of these are separate forces in the framework. They are different views of the same resolution-seeking process operating on configurations at different scales and through different bonding modes. The unification is achieved not by combining four force fields into one but by recognizing that the four "forces" were always the same geometric process viewed through different experimental windows.

12.3 Energy

Energy in the framework is the interaction event itself. When two configurations interact — when an outer corner shares a coordinate with another configuration's central junction — the bonding event registers as an energy exchange. The energy is the actual hit, the moment of geometric alignment, not a separate substance that the configurations carry between them.

This reframing has consequences for how energy conservation works in the framework. Energy is not a conserved scalar that flows from one configuration to another like a fluid. Energy is the count of bonding events and their associated tilt costs. Total energy conservation across an interaction corresponds to the conservation of total content across the configurations involved (three X values, three O values, three Y values per configuration, preserved through all axis-tilt operations and bonding events).

The full development of energy as the interaction-event count, including its relationship to standard physics' kinetic and potential energy formulations, is developed in Paper 8 (mass and energy accounting) and Paper 9 (heat as the interaction event).

12.4 Momentum

Momentum in the framework is carried by configurations through their motion through the interaction fabric. A photon at c carries momentum corresponding to its propagation velocity. A mass configuration at below- c velocity carries momentum corresponding to its velocity and its accumulated tilt cost (which the framework identifies as its mass).

Momentum exchange in collisions corresponds to bonding events that redistribute tilt across the participating configurations. When two configurations collide and rearrange their bonds, the change in their post-collision velocities relative to their pre-collision velocities reflects the redistribution of tilt across the new bonding configuration.

The full development of momentum as the framework's reformulation of motion through the fabric, including its relationship to the standard physics momentum vector, is developed in Paper 7 (acceleration, momentum, and geometric consolidation).

12.5 Time

Time in the framework is the ledger of the 720° rotation pattern's progress for tilted configurations. The photon at zero tilt does not need to rotate to maintain its resolution, so it has no rotation ledger and no proper time registers. Mass configurations have accumulated tilt that requires the 720° rotation pattern to maintain resolution (developed in Appendix B), and time registers for them as the rotation progresses through the ledger.

This is why photons experience no proper time — their geometric structure does not require the rotation ledger that other configurations do. It is why time dilation registers naturally for relativistic configurations — their rotation ledger runs at a different rate compared to slower observers. The macroscopic arrow of time is

the integrated sum of these individual 720° rotation ledgers, which are themselves driven at the micro-scale by the relentless, non-invertible cadence of the temporal scan, each cycle marked distinct by a_k . The full development of this comparative-ledger unification is the work of Paper 13.

The framework establishes a strict asymmetry in temporal registration: time is an operational consequence exclusive to configurations carrying mass-tilt, driven by the execution of the temporal scan. Conversely, the photon baseline (State 9) carries zero tilt, runs no temporal scan, and consequently does not register time. A massive configuration decaying back into the photon baseline does not constitute a temporal reversal; it represents a forward geometric transition in which the local time-registering mechanism is deactivated. The universal sequence ledger, with each entry marked distinct by a_k , remains strictly cumulative and non-invertible, permanently recording the historical trajectory.

The full development of time as the rotation ledger, including the comparative-ledger account of relativistic time dilation and the framework's treatment of the arrow of time, is developed in Paper 13.

12.6 Probability and Apparent Randomness

A framework built on a discrete deterministic manifold must account for the probabilistic structure that quantum mechanics observes. The framework's position is that it operates on geometric determinism with a complex background density, and that quantum probability is an emergent appearance rather than fundamental randomness.

The mechanism is as follows. A configuration is never traveling through a truly empty vacuum. As established in Section 3.2, the fabric is the continuous three-axis coordinate manifold, everywhere populated with shifting open background states (States 2 through 8) constantly rearranging as they relieve their structural tension, along with the balancing tilts that maintain local equilibrium. A traveling configuration navigates this dense, constantly-updating sea of background geometry.

Consider a pinball analogy. The ball follows strict deterministic laws of motion, but because the board is covered with many small moving bumpers, the path looks random to a distant observer who cannot track the bumpers. In the framework, the "bumpers" are the shifting open background states whose precise positions at the discrete a_k scale are not tracked by standard instruments. Standard physics observes the unpredictable path, cannot resolve the background structure causing it, and describes the result as quantum probability. The framework's claim is that the underlying process is deterministic at the manifold level, and the apparent randomness reflects the unresolved complexity of the background density rather than fundamental indeterminacy.

What standard physics calls an "interaction rate" or "coupling constant" is, in this reading, a measure of geometric alignment frequency. Whether a traveling configuration interacts with another depends on whether their discrete a_k coordinates overlap during passage. Because configurations and background states are constantly shifting their axis tilts, the interaction "probability" is the geometric fraction of time that the structures happen to align — a spatial-temporal synchronization frequency rather than a fundamental dice roll.

The framework does not claim to have derived the full quantum-mechanical probability distributions from this picture; that quantitative work belongs to later papers. What it offers here is a structural account of where apparent randomness comes from — emergent from unresolved background density on a deterministic manifold — rather than treating randomness as a fundamental primitive. The conventional quantum phenomena usually cited as evidence of fundamental randomness or non-locality are examined in the critique companion, Paper 2.5, where they are read as artifacts of apparatus and statistical method rather than as mysteries.

13. Charge as Axial Orientation

A framework built on three rotational axes and three triangles meeting at a central junction has to account for charge directly. In this framework, charge is not an intrinsic property pinned to a dimensionless point; it

is a structural consequence of the same three rotational axes that organize the twenty-seven configurations. This section develops that account. The question of whether the conventional treatment of charge — as a primary property inherited from the matrix structure of the Dirac equation — was ever derived or simply adopted is taken up directly in Paper 2.5, the first critique paper of this series; here the focus is on building the framework's own positive account.

13.1 Charge Reframed: The Direction of Unwinding

Within the framework's geometry, what standard physics labels as charge is reframed as the directional orientation a configuration takes when it tries to unwind back toward the photon baseline. The same axis-tilt process that produces mass also determines the unwinding direction.

The electron (State 1 with central junction (Y, Y, Y)) and the positron (State 1 with central junction (X, X, X)) are at structurally equal distances from the photon baseline. There is no preferred direction in the framework's geometry. What standard physics records as "positive charge" or "negative charge" is the framework's account of which direction a configuration moves when an external field forces it to unwind toward the photon state.

13.2 Asymmetric Paths from Biased Configurations

Configurations in States 2, 3, 5, 6, 7, and 8 carry structural asymmetries at their central junctions. A State 7 configuration with (X, O, O) at center is closer to the positron-direction tilt; a State 8 configuration with (Y, O, O) at center is closer to the electron-direction tilt. These structural biases determine which path each configuration follows when an external interaction forces unwinding.

13.3 Fractional Quark Charges Explained

Standard physics records quarks with fractional charges ($+2/3$ and $-1/3$ in units of the elementary charge). The framework's configuration values are the discrete structural identifiers X, O, and Y, not fractions. At first glance these appear incompatible. They are not.

The framework's account: quarks (single-axis configurations from States 7 and 8) never exist as independent particles because their open structure cannot satisfy resolution on its own. They are observed only as constituents of resolved configurations (three-node resolution) or two-link open configurations (transient bonds). When standard physics measures the "charge" of an individual quark within a resolved configuration, it is not measuring an intrinsic property of a part — it is probing a tri-configuration bond and resolving only one of its three nodes at a time. The probe touches one node of a locked three-node structure and registers one-third of the whole. That one-third is not a fractional charge carried by a fundamental particle; it is a partial resolution of a single whole geometric bond. The fraction is an artifact of measuring one node of three, not evidence of a thing that owns one-third of a charge.

The history of the fractions, and why they were never fundamental. It is worth saying plainly where the fractional charges came from, because the history shows they were a bookkeeping device, not a discovery. In the 1960s, classifying particles by the observed symmetry of the "Eightfold Way," physicists found that protons and neutrons would not fit a simple integer-based model. To make the arithmetic balance — to recover the measured proton charge of $+1$ and neutron charge of 0 — they introduced quarks and assigned them fractional charges of $-1/3$, $+1/3$, and $+2/3$. These fractions were not found to be fundamental; they were chosen because they were the values that, summed in threes, reproduced the macroscopic results already measured. The arithmetic was forced to fit the observation rather than read off the geometry.

And here is the irony the framework cannot let pass: every one of those charges is built on a denominator of three. The whole apparatus depends on dividing by three, and a thing measured in three parts. The Standard Model spent sixty years constructing color charges and flavor physics to explain why the numbers come in

thirds — never recognizing that the thirds are there because they are probing a tri-configuration bond and touching one of its three nodes at a time. The fraction exists solely because the measurement counts one part of a three-part convergence. Had the three-part structure of the bond been recognized, no fractional charge would ever have been needed: the geometry is a three-node system, and the probe was simply registering one node of three. The framework reads the same number cleanly — one of three — where the conventional account reads a mysterious fundamental fraction. They have been looking at the Rule of Three the entire time and calling it a mystery.

13.4 Omnidirectional Fabric Interaction, Proximity Scaling, and the One-Way Arrow of Time

The framework establishes a strictly localized, omnidirectional expansion mechanic across the continuous, unpixelated fabric manifold, completely eliminating macroscopic action-at-a-distance and any inherent spatial directionality. A configuration does not communicate through restricted linear paths or step-by-step sequential channels. Instead, during its temporal scan, the configuration balloons or foams outward symmetrically in every direction of the surrounding geometry simultaneously, strictly bounded within its immediate contiguous coordinate neighborhood. This omnidirectional sweep evaluates the entire local micro-environment in parallel. Consequently, a single scanning configuration can interface with multiple distinct surrounding states—such as a State 7 and a State 8 configuration occupying directly adjacent neighbor coordinates—at the exact same moment. This parallel audit operates entirely within this localized boundary step; it carries no capacity for non-local propagation or instantaneous signaling across macroscopic distances, preserving absolute physical locality.

Crucially, a strict distinction must be maintained between spatial isotropy and temporal asymmetry: while the fabric matrix possesses zero directional bias across the dimensions of space, the sequential accumulation of these omnidirectional execution cycles introduces a strict, non-invertible progression in the dimension of time. This non-directional spatial interaction provides the direct mechanical foundation for macroscopically observed temporal phenomena:

1. The One-Way Arrow of Time: The macroscopic unidirectional flow of time is derived directly from the forward-accumulating nature of the geometric resolution ledger across the surrounding, unpixelated fabric units. As long as a configuration remains in an off-baseline state (States 1–8), its temporal scan must process local coordinate tension, ticking the ledger forward. A configuration can absolutely resolve its structural imbalance and return to the State 9 photon baseline. However, returning to the photon floor does not run the clock in reverse. Instead, upon transitioning into State 9, the configuration drops the temporal scan, running only its identity scan. Because the temporal scan is omitted at the baseline, no time is ever registered at State 9. The arrow of time is strictly one-way because it accumulates forward during off-baseline processing, and exits into a timeless state upon photon resolution.

2. Proximity Scaling of Forces: This omnidirectional fabric mechanic also provides the direct geometric origin of inverse-distance force scaling (the macroscopic observation of repulsion increasing at closer range):

- **Vast Separation:** When two identical off-baseline configurations (such as two electrons) are separated by a significant distance, the non-resolution conflict is distributed across an immense number of surrounding fabric units. The structural tension is widely buffered and absorbed by the manifold, resulting in a minimal localized coordinate gradient.
- **Proximal Compression:** As the distance decreases, fewer surrounding geometric units remain between the cores to evaluate and distribute the structural baseline deviation. The absolute failure to achieve a State 9 photon baseline resolution is compressed into a dwindling number of fabric cells. This lack of surrounding geometry causes an immediate, severe compression spike in local coordinate tension, forcing the configurations apart with maximum mechanical velocity.

The screening valuation that drives this interaction has three outcomes during the temporal scan:

Attraction (the resolution match). If the neighboring configuration holds a complementary value that settles the open junction (an unresolved X-tilt meeting the Y-tilt that completes its triplet, for instance), the coordinate tension drops and the configurations pull together. This geometric optimization is what is macroscopically observed as electrostatic or nuclear attraction.

Repulsion (the non-resolution push). If the neighbor holds a conflicting value (for instance, matching integers that are already resolved, or values that cannot settle the open junction), bonding with it would aggravate the imbalance and push the system even further away from the State 9 photon baseline. The system rejects the interaction, mechanically forcing the unhelpful configuration away. This structural defense mechanism is what we observe as repulsion.

No interaction (stays neutral). If the neighbor offers neither a resolving match nor a conflicting obstruction, nothing passes between them and the configuration remains as it is. The framework has no isolated state — the integers are the fabric, so a configuration is always embedded in surrounding geometry — and this is simply the neutral case, where that surrounding geometry neither completes nor compounds the configuration's tension.

11.5.1 Why Identical Cores Repel: A Destination Failure, Not an Identity Rejection

A clarification prevents a misreading that would otherwise contradict the bonding rules of Section 10. In Section 10, an outer value docks into a center holding two matching values — for instance, a third Y completing a core toward saturation. This means matching values do bond in that context. So why do two electrons (each a saturated all-Y core) repel rather than bond?

The answer is that repulsion is governed by destination, not by sign-matching. When two already-saturated, resolved configurations interface, each runs its local temporal scan and asks whether bonding with the neighbor would shift its coordinates toward the flat State 9 baseline. Because the neighbor is an identical, already-resolved configuration, bonding offers no shift toward baseline — there is nothing to resolve, no pathway forward, a complete dead-end. The interaction is rejected, and that rejection manifests as repulsion. The repulsion is not caused by the two configurations carrying the same value; it is caused by the encounter providing no route toward resolution. By contrast, an open center holding two matching values (such as a partial core needing one more) genuinely welcomes a third matching value, because that docking completes a saturation step and advances the local geometry. Matching values bond when the bond advances resolution and repel when it offers only a dead-end. The governing variable is always the path toward the baseline, never the bare sign of the value.

13.5 The Central Junction's Structural Immunity to Internal Repulsion

A foundational distinction must be maintained between the framework's primary matrix values and the emergent phenomenon of electromagnetic repulsion. A conventional critique examining configurations that carry multiple identical nonzero elements at the central junction—such as the two Y values in State 6 (Y, Y, O) or the three Y values in the self-contained electron configuration (Y, Y, Y)—might assume these co-located states should exert a repulsive force on one another, destabilizing the central vertex.

This objection represents a category error. The central junction is not an open region of space where independent fields interact via point-charge dynamics. Instead, the vertex is the singular common intersection point of three independent, orthogonal rotational planes representing Axis 1, Axis 2, and Axis 3. The Y or X values registered at the center are static structural integer values defining the geometric axis-tilt state of each native triangle relative to the baseline fabric.

Because repulsion is strictly an emergent mechanical resolution driven by local neighborhood screening during a configuration's temporal scan, it has no operational mechanism *internal* to a single configuration's core. The identical integer values do not interact as external force carriers; they simply coexist as the structural

definition of the local coordinate intersection. They are integer values, not colliding charges, and they possess complete structural immunity to internal repulsion.

13.6 What the Charge Account Establishes

The framework treats charge as a structural consequence of its axis-spin geometry rather than as an intrinsic property of point particles. Neutrality is native to the three-triangle structure: the O value sits in the same matrix as the X and Y values, with no separate apparatus required for neutral entities. This is the framework's positive account, and it stands on its own geometric terms.

It naturally raises a question about the conventional account, which builds charge in as a primary category and then constructs separate field theories for every neutral entity. Was that primacy of charge ever derived, or was it adopted because the mathematics of the Dirac equation made it convenient? That question — and the broader pattern it belongs to — is the subject of Paper 2.5, the first critique paper of this series. Here it is enough to establish that the framework's geometry does not need charge as a primitive; it produces charge, neutrality, and fractional values from one structure.

14. The Vocabulary Established for the Series

Every subsequent paper in this series uses the geometric structure established here. The framework's account of mass in Paper 8 measures mass as the configuration's accumulated departure from the photon baseline through axis tilts. The framework's account of gravity in Paper 10 treats gravitational effects as the fabric's response to resolved configurations' outer bonds searching for resolution toward c. The framework's account of cosmic expansion in Paper 14 traces expansion to the same outer-bond search integrated across all mass in the universe. The framework's account of time in Paper 13 traces time to the rotation patterns of tilted configurations.

The notation introduced here — ordered triplets for central junction content, the nine-state classification, and the three-triangle geometric structure — is the framework's compact way of referring to specific configurations throughout the series.

15. Systemic Architecture Roadmap

The geometric foundation established in this paper provides the complete structural grammar for the framework. To maintain analytical focus, the quantitative derivations of specific particle masses, decay rates, and interaction cross-sections are deliberately distributed across the subsequent treatises in this coordinated series. This is an architectural choice of modular isolation, ensuring each physical phenomenon is developed with full rigor in its own dedicated treatment rather than crowded into the foundational paper. Paper 2 defines the geometry; the subsequent papers deploy it.

The deployment schedule across the series is established as follows:

Paper 4 develops the detailed dynamics of axis-tilt operations and electron-positron pair production from photons.

Paper 5 develops the full resolution analysis determining compositional stability across multi-node networks.

Paper 8 develops the quantitative mass derivation as a departure function from the photon baseline, defining the single-axis, open, and resolved configuration spectrum in specific values.

Paper 10 develops gravitational mechanics derived from the fabric's integrated accounting of resolved-configuration resolution-seeking.

Paper 13 develops temporal mechanics via the 720° rotation ledger.

Paper 14 develops cosmic expansion metrics mapped as the integrated outer-bond search, including the integration mechanism that bridges resolved-configuration-scale dynamics to the observed expansion history.

Paper 15 develops beta decay and antimatter as tilt-direction phenomena.

Paper 16 develops the complete configuration compositions for open, resolved, and over-constrained configurations, matching observed experimental states across the particle spectrum.

The thermodynamic foundations of the framework, including its account of entropy, are identified as active research directions within the series rather than as completed derivations. The framework states this honestly as the frontier of its current development.

This modular structure means that Paper 2 should be judged on whether its geometry is sound and internally consistent, not on whether it contains the quantitative mass values or decay rates that belong to the later treatises. Each phenomenon has its dedicated treatment; the foundation laid here is what they all build upon.

16. Summary

The framework rests on three rotational axes organizing the interaction fabric, each carrying one of three discrete values: X, Y, and O, which are tilt directions and a neutral anchor, not signs or quantities. A configuration is a three-triangle structure meeting at a central junction, occupying nine positions whose content is always three X, three Y, and three O.

The photon (State 9) is the parent state: zero tilt, no mass, propagating at c . Mass is departure from that baseline through axis-tilt operations — more tilt, more mass, slower than c — and the kinematic gap holds that self-contained mass configurations can neither reach c nor rest at $v = 0$. a_k is the identifier marking each transition distinct from the next.

The twenty-seven configurations partition into nine categories by central-junction signature. Only two are intrinsically resolved and stable alone — State 9 (photon) and State 1 (electron-positron); the other seven are open and require bonding. State 4 is distinctive only in that three of them bonding at a shared center produce the perfect three-axis balance from which a heavy X-dominant configuration, an electron, and a photon resolve; it sits close to the photon baseline, not far from it.

Bonding proceeds by outer-to-center docking and outer-to-outer convergence, and the chain is named by geometry: dual-axis-plus-one-surplus (first link), tri-axis-plus-one-surplus (transient, conventionally the meson), triple tri-axis (resolution, conventionally the baryon), with tri-axis-plus-two-surplus (conventionally tetraquarks) forming briefly when surplus exceeds what resolution can hold. Resolved configurations resolve internally while their outer bonds keep searching toward the photon state — the structural origin of both gravity and cosmic expansion.

Charge is the directional axial orientation of unwinding back toward the photon baseline, not an intrinsic stamp; fractional quark charges are shared-geometry effects within three-node resolved configurations. There is no neutrino in the framework: what is called one is a collision product, the geometry produced by whatever made the hit.

One appendix follows: Appendix B derives the 720° rotation signature of mass configurations. The critique companion (Paper 2.5) examines the Dirac formalism, the beta-decay experiments, the so-called neutrino, the nuclear interior, and collider spectra.

The next paper in this series, Paper 3, develops the inside-outside dynamics in greater detail and explains the rotation operations that move values between central junction and outer corner positions, including the geometric mechanism behind electron-positron pair production from photon interactions.

17. Falsifiability

A framework is only scientific if it can be wrong. This section states the conditions under which the framework presented in this paper would be falsified, so that the reader can see the framework is making checkable commitments rather than unfalsifiable claims.

The framework's geometric foundation would be falsified by any of the following observations:

The discovery of a stable, independently-propagating particle that cannot be placed within the twenty-seven-configuration structure or built as a compositional network of those configurations would falsify the

enumeration. The framework commits to the claim that the configuration space is exactly twenty-seven and that all observed particles are these configurations or their compositions.

The discovery of a stable single particle carrying a configuration from States 2 through 8 (an open configuration) propagating freely without bonding would falsify the resolution principle. The framework commits to the claim that only State 9 (photon) and State 1 (electron-positron) are intrinsically resolved and stable on their own without bonding, and that all other configurations require compositional resolution.

The measurement of an isolated free quark with a fractional charge propagating independently would falsify the framework's account of fractional charge as a shared-geometry effect. The framework commits to the claim that fractional charge arises only within multi-node bonded structures and that single quarks never propagate alone.

The observation of energy-dependent variation in the speed of light in vacuum (photon dispersion) attributable to spacetime discreteness would challenge the framework's account of the photon baseline propagating at c regardless of energy. The framework commits to the claim that its discreteness does not produce such dispersion.

A demonstration that the 720° rotation signature of fermions does not follow from the internal-then-external resolution sequence — for example, the discovery of a massive configuration that returns to its starting state after a single 360° rotation, or a massless one requiring 720° — would falsify the framework's account of spin.

A demonstration that mass does not scale with departure from the photon baseline — for example, the discovery of a heavily-bonded composite that propagates at or above the velocity of a lighter configuration with all else equal — would falsify the framework's account of mass as accumulated departure from the photon baseline.

Two of the framework's structural commitments are tied to existing quantitative evidence and are listed here as falsification conditions, with their full development reserved for the critique and phenomenological papers. First, the reading of the conventional particle count as geometric duplication (matter-antimatter as tilt direction, color as rotational phase-shift) is tied to the empirical R-ratio in electron-positron annihilation: if the framework's geometry cannot reproduce the measured R-ratio scaling (read as the three axes of the fabric rather than three color charges), the twenty-seven-configuration reading of the spectrum is falsified. The historical and methodological case is made in Paper 2.5 (Section 7.1); the quantitative map is developed in Paper 16. Second, the framework's claim that what is called the neutrino is a collision-produced geometry rather than a travelling particle is falsified if the recorded detection events cannot be reproduced as products of collision geometry — that is, if the data demands a persistent travelling entity that the framework denies. This is confronted against the detection and oscillation data in Paper 2.5 and Paper 12. Both falsification conditions are held as genuine, not settled results.

These are stated as structural falsification conditions. The quantitative falsification conditions (specific predicted mass values, decay rates, and cross-sections that could be checked against measurement) are developed in the focused papers that derive those quantities, particularly Paper 8 (mass spectrum) and Paper 16 (particle compositions). The framework's commitment is that it makes checkable predictions at both the structural level stated here and the quantitative level developed downstream.

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19. Appendix B: Derivation of the 720° vs. 360° Rotation

This appendix derives the framework's account of why off-baseline configurations require 720° rotation to return to their starting state while the photon (State 9) requires only 360° . The derivation is presented first as physical intuition grounded in the configuration geometry, then as a mechanical derivation from the framework's resolution-seeking principle. The derivation deliberately does not proceed through a rotation matrix on the three axes, for reasons made explicit in B.2: the spinor sign that distinguishes fermions lives in the double cover of the rotation group and cannot be honestly reproduced by any operation on three real spatial axes. The framework instead grounds the 720° signature in its own native mechanics.

A clarifying note on which configurations the 720° applies to, and what the second sweep does for each. State 9 alone is the zero-tilt photon baseline; it is the only configuration that requires no second sweep,

completing its single 360° internal check and propagating at c . Every configuration that has departed from the baseline — States 1 through 8 — is mechanically required to run the full 720° two-sweep audit. What differs between them is the purpose the second sweep serves, which depends on whether the configuration is open or resolved:

For the open configurations (States 2 through 8), the second 360° sweep is an active resolution search: the configuration scans its immediate neighborhood for the complementary values that would let it bond toward resolution, or, failing that, reconciles by shedding its surplus toward the baseline. These are the states whose unresolved outer values drive the outward resolution search that the framework connects to cosmic expansion (Section 10.4).

For the resolved configuration State 1 (electron-positron), the second 360° sweep is a baseline-orientation check rather than a resolution search. State 1's internal core is already saturated and resolved, so the first sweep registers internal resolution — it is stable matter and does not need to find a bonding partner to complete itself. But because State 1 has still departed from the baseline (it carries mass), it must still run the second sweep, and that sweep registers its standing relative to the broader fabric: "I am not a photon; my core is resolved, so I can persist as stable matter for now, but I remain an off-baseline configuration whose ultimate resolution is a return to the State 9 floor." This is why State 1 exhibits the full 720° spinor signature that standard physics observes for the electron — it runs both sweeps — while not contributing to the outward resolution search that drives expansion, because its second sweep confirms resolution rather than seeking it. The universal rule is therefore clean: if a configuration has mass (States 1 through 8), it runs the full 720° cycle; only the massless baseline photon (State 9) runs a single 360° .

19.1 B.1 The Physical Intuition

Standard physics treats spin as an abstract mathematical point-property. The framework grounds the 720° versus 360° distinction in the difference between a resolved configuration and one carrying unresolved geometry. The photon, resolved at the baseline, needs one sweep; a mass configuration, carrying unresolved tilt, needs two.

The photon (State 9) at 360° . The photon configuration carries zero tilt at its central junction. Its three triangles are in their lowest-cost arrangement: O at center, X and Y balanced at outer corners. There are no internal twists, no localized constraints, no mass displacements. The configuration travels along the spatial baseline at c .

Rotating a straight, open, fully-resolved structure around its axis of travel returns it to its exact initial orientation after a single complete turn. Because the photon has no internal tilt and no unresolved geometry to assess, it requires only a 360° sweep to match its starting state. One complete turn, one complete return.

The electron-positron pair (State 1) at 720° . The moment tilt is introduced to drop a configuration below c and generate mass, the configuration acquires unresolved geometry that must be accounted for. To produce a State 1 configuration, three axis-tilt operations move the X (or Y) saturation values from the outer corners into the central junction, forming a self-contained resolved structure that has departed from the baseline. The configuration now carries structural tension that a single sweep cannot fully evaluate.

The intuition, made mechanical in B.2, is that an off-baseline configuration must assess two things — its internal departure from the baseline, and its external prospects for resolution against the surrounding fabric — and that these two assessments require two complete sweeps. The configuration returns to its true starting state only after both sweeps are complete, for a total of 720° .

The 720° requirement is therefore not an arbitrary mathematical feature of spinor algebra. It is a direct consequence of the configuration having departed from the baseline and acquired unresolved geometry that requires a two-phase evaluation.

19.2 B.2 The Mechanical Derivation: The Double-Sweep Resolution Check

The physical intuition above can be made precise without borrowing the $SU(2)$ matrix from standard physics and without any algebraic sleight of hand. The 720° requirement follows directly from the framework's own law of resolution-seeking: a configuration that has departed from the photon baseline must perform two complete geometric sweeps to fully evaluate its structural state, while a configuration already at the baseline needs only one.

It is worth stating plainly why this derivation does not proceed through a rotation matrix on the three axes. A full 360° rotation in ordinary three-dimensional space is, group-theoretically, the identity operation — any 3×3 real rotation matrix for 2π is the identity, with determinant $+1$. The famous minus sign that distinguishes spin-1/2 particles lives in $SU(2)$, the double cover of the rotation group, and it cannot be reproduced by any single operation on three real spatial axes. A framework that claimed to derive the spinor sign from a 3×3 axis permutation would be making a checkable error, since the cyclic permutation of three axes is an even permutation with determinant $+1$, not -1 . The framework does not make that claim. Instead, it derives the 720° signature from the mechanical logic of resolution-seeking, which is the framework's own native principle.

The photon (State 9): 360° , already resolved. The photon baseline sits perfectly flat on the central axis with zero tilt and zero departure from the baseline. Because it is completely resolved, it carries no structural tension. There is nothing for it to assess and nothing for it to seek. A single 360° geometric sweep returns it to its starting state because it has nothing to look for. One sweep, one return.

Off-baseline configurations: 720° , resolution-seeking. The moment a configuration tilts off the central axis and departs from the baseline, it generates unresolved geometry. The law of conservation that governs the framework demands that the configuration constantly seek resolution to balance that structural tension (Section 5.2). Because the configuration is tilted and carrying unresolved geometry, a single 360° sweep is not sufficient to evaluate its complete structural state. The configuration is mechanically required to perform a double sweep totaling 720° , in two distinct phases:

Sweep 1 — the internal identity check (first 360°). The configuration first sweeps its own internal matrix to determine what it is. This internal check resolves a three-way question. Is the configuration the photon (State 9, the flat baseline)? If so, it is fully resolved and stops at 360° — this is the only outcome that completes at a single sweep. Is it a self-contained State 1 (the saturated electron-positron, internally resolved)? If so, its core is complete and it is stable matter, but it has still departed from the baseline, so it proceeds to a second sweep that serves as a baseline-orientation check rather than a resolution search. Or is it one of the open States 2 through 8? If so, the internal check registers that the configuration is open, carrying unresolved outer values, and it proceeds to a second sweep that serves as an active resolution search. Both the resolved State 1 and the open States 2 through 8 — every configuration that has left the baseline — proceed to the second sweep; only the baseline photon stops at one.

Sweep 2 — the external resolution scan (second 360°). Once the internal check has registered that the configuration is an open state (States 2 through 8) with unresolved outer values, the configuration uses the second full turn to scan its outer boundary against the surrounding fabric, looking for matching open geometry to bond with and achieve resolution. The outer values are searching for the partners that will complete their resolution. This outward search is governed by conservation: the unresolved outer values must find resolution, and that conserved outward search is mechanically real.

Only after completing both the internal identity check and the external resolution scan — 720° in total — has the off-baseline open configuration fully evaluated its structural status and returned to its true starting state. The doubling is not a mathematical accident or an imported postulate. It is a mechanical necessity: an open configuration carries unresolved outer values, and evaluating them requires both an internal identity check and an external resolution scan, which is two complete sweeps.

The reason a single sweep cannot accomplish both tasks at once is a strict logical dependency, not an arbitrary stipulation. The external scan depends entirely on the output of the internal check. A configuration cannot evaluate its standing against its surroundings until it has first confirmed what it is — whether it is the resolved photon (State 9), an internally-resolved State 1, or an open state (States 2 through 8). This is the same structure as a conditional statement in a computation: the body of a clause cannot execute before the condition it depends on has been evaluated. The first 360° sweep evaluates the condition (am I the baseline photon, resolved matter, or open?). Only State 9 returns "baseline photon" and stops at 360° . Every other result — resolved State 1 or open States 2 through 8 — triggers the mandatory second sweep. For an open state, the second sweep is the resolution search (scan the neighborhood for resolution partners). For State 1, the second sweep is the baseline-orientation check (confirm resolution and register standing relative to the fabric). Either way, the second sweep cannot run before the first has determined which case applies. This sequential dependency is what forces the two-stage structure and produces the 720° signature for every off-baseline configuration.

19.3 B.3 What This Derivation Accomplishes

The derivation produces the 720° signature from the framework's own resolution-seeking principle rather than from borrowed group theory or a false matrix identity. The photon, fully resolved at the baseline, needs only one sweep because it has nothing to assess or seek. The mass configuration, carrying unresolved tilt, needs two sweeps — internal check plus external scan — because resolution-seeking is a two-phase mechanical process for any off-baseline configuration.

This gives the reviewer a concrete, algorithmic reason for the spinor rotation signature that aligns exactly with the framework's resolution-seeking mechanism established in Section 5. The 720° signature is not presented as a mystical mathematical property and not as a group-theory limitation smuggled in from outside. It is a rigid mechanical necessity for any configuration that has been knocked off the baseline and must evaluate both its internal departure and its external resolution prospects before its structural state is fully defined.

The framework's account is consistent with the standard physics observation that fermions require 720° while bosons (like the photon) require 360° . Standard physics describes this through the $SU(2)$ double cover of the rotation group; the framework provides a mechanical interpretation of why the doubling occurs physically — the two-phase resolution check required by any off-baseline configuration. The framework does not contradict the $SU(2)$ description; it offers a structural reading of what the double cover corresponds to in the framework's geometry. The full formal relationship between the double-sweep resolution check and the $SU(2)$ double cover is an active research direction within the series; what the framework establishes here is the mechanical necessity of the doubling from its own principles, which is sufficient to reproduce the observed $360^\circ/720^\circ$ distinction.

19.4 B.4 Why This Derivation Appears in Paper 2

The 720° rotation signature is a fundamental feature of fermions in standard physics. Any framework that introduces a new geometric structure for matter configurations must either reproduce the 720° signature or explain why it does not appear. The framework presented in this series reproduces the signature naturally as a consequence of the three-triangle geometry and the axis-tilt mass-generation mechanism. This appendix demonstrates that reproduction.

Subsequent papers in the series develop the consequences. Time as the 720° rotation ledger (Paper 13) builds directly on the derivation here. The framework's treatment of spinor algebra in particle interactions (Paper 8 and subsequent papers) uses the geometric foundation established in this appendix.

End of Paper 2 draft.